Thinking causally

Estimands, DAGs, and model specification

Sebastian Ramirez-Ruiz
European University Institute (EUI)
October 16, 2026



📇 I am Sebastian Ramirez-Ruiz, or just Sebastian.

🎓 Part-time Assistant Professor in Quantitative Methods and Max Weber Fellow (Department of Political and Social Sciences)

📧 sebastian.ramirezruiz@eui.eu



I am Sebastian Ramirez-Ruiz, or just Sebastian.

Part-time Assistant Professor in Quantitative Methods and Max Weber Fellow (Department of Political and Social Sciences)

sebastian.ramirezruiz@eui.eu

I am a **computational social scientist**



📇 I am Sebastian Ramirez-Ruiz, or just Sebastian.

🎓 Part-time Assistant Professor in Quantitative Methods and Max Weber Fellow (Department of Political and Social Sciences)

📧 sebastian.ramirezruiz@eui.eu

I am a **computational social scientist** researching how **political decisionmakers** engage with **evidence** and **information**,



📇 I am Sebastian Ramirez-Ruiz, or just Sebastian.

🎓 Part-time Assistant Professor in Quantitative Methods and Max Weber Fellow (Department of Political and Social Sciences)

✉ sebastian.ramirezruiz@eui.eu

I am a **computational social scientist** researching how **political decisionmakers** engage with **evidence** and **information**, and how these interactions shape policy and governance across contexts.

Two questions before any estimation

Two questions before any estimation

With Kevin, you asked **what a regression is for**: description, prediction, or causal inference.

Two questions before any estimation

With Kevin, you asked **what a regression is for**: description, prediction, or causal inference.

Week 3 (today)

What exactly are we trying to estimate?

Two questions before any estimation

With Kevin, you asked **what a regression is for**: description, prediction, or causal inference.

Week 3 (today)

What exactly are we trying to estimate?

Weeks 4–5

Do our numbers measure the concepts we care about?

Two questions before any estimation

With Kevin, you asked **what a regression is for**: description, prediction, or causal inference.

Week 3 (today)

What exactly are we trying to estimate?

Weeks 4–5

Do our numbers measure the concepts we care about?

Both questions come **before** you choose a model.

This session

This session

Thinking causally

Causation and causal reasoning, counterfactuals, potential outcomes, and the biases of naive comparisons

This session

Thinking causally

Causation and causal reasoning, counterfactuals, potential outcomes, and the biases of naive comparisons

What is your estimand?

Unit-specific quantity and target population, stated before any model

This session

Thinking causally

Causation and causal reasoning, counterfactuals, potential outcomes, and the biases of naive comparisons

What is your estimand?

Unit-specific quantity and target population, stated before any model

DAGs and good and bad controls

Confounders, mediators, colliders, the backdoor criterion, and why more controls is not safer

This session

Thinking causally

Causation and causal reasoning, counterfactuals, potential outcomes, and the biases of naive comparisons

What is your estimand?

Unit-specific quantity and target population, stated before any model

DAGs and good and bad controls

Confounders, mediators, colliders, the backdoor criterion, and why more controls is not safer

Where do your data come from?

Knox, Lowe & Mummolo (2020) in depth: selection, data-generating processes, and what records can and cannot tell us

This session

Thinking causally

Causation and causal reasoning, counterfactuals, potential outcomes, and the biases of naive comparisons

What is your estimand?

Unit-specific quantity and target population, stated before any model

DAGs and good and bad controls

Confounders, mediators, colliders, the backdoor criterion, and why more controls is not safer

Where do your data come from?

Knox, Lowe & Mummolo (2020) in depth: selection, data-generating processes, and what records can and cannot tell us

Practice

Some group discussion today; DAGs and simulations in R in Lab 3 on Monday

Why causation matters

Why causation matters

Why causation matters

Causation: a poor man's working definition

- a.k.a. causality, cause and effect
- The idea that one variable [event, process, object; the cause] leads to another variable [event, process, object, outcome; the effect]

Why causation matters

Causation: a poor man's working definition

- a.k.a. causality, cause and effect
- The idea that one variable [event, process, object; the cause] leads to another variable [event, process, object, outcome; the effect]

Causal reasoning

- The process of studying and identifying **causality** (*ruling out non-causal sources of association*)
- **Causal inference** is one way of doing causal reasoning

Why causation matters

Causation: a poor man's working definition

- a.k.a. causality, cause and effect
- The idea that one variable [event, process, object; the **cause**] leads to another variable [event, process, object, outcome; the **effect**]

Causal reasoning

- The process of studying and identifying **causality** (*ruling out non-causal sources of association*)
- **Causal inference** is one way of doing causal reasoning

> **Side note: what can be a cause?** The **manipulationist** camp holds there is **no causation without manipulation** (Holland 1986): only things we could, in principle, intervene on are causes, so "the effect of race" or "the effect of sex" is *not a thing* for them. The **difference-making** (or structural) camp counts anything that makes a difference to the outcome. Others break attributes into manipulable parts (Sen & Wasow 2016: race as a "bundle of sticks"). That is why we start with a loose, working definition and get precise once we define potential outcomes.

Why causation matters for your thesis

Why causation matters for your thesis

If you are here, my hunch is...

Why causation matters for your thesis

If you are here, my hunch is…

- you are writing an empirical research paper or thesis

Why causation matters for your thesis

If you are here, my hunch is…

- you are writing an empirical research paper or thesis
- you are interested in the effect of *X* on *Y*

Why causation matters for your thesis

If you are here, my hunch is…

- you are writing an empirical research paper or thesis
- you are interested in the effect of *X* on *Y*
- you use observational data, e.g., survey data or time-series cross-sectional data, and have lots of information about your cases

Why causation matters for your thesis

If you are here, my hunch is…

- you are writing an empirical research paper or thesis
- you are interested in the effect of *X* on *Y*
- you use observational data, e.g., survey data or time-series cross-sectional data, and have lots of information about your cases
- you are asking: **should I control for *Z*?** Could it actually introduce problems?

Why causation matters for your thesis

If you are here, my hunch is…

- you are writing an empirical research paper or thesis
- you are interested in the effect of *X* on *Y*
- you use observational data, e.g., survey data or time-series cross-sectional data, and have lots of information about your cases
- you are asking: **should I control for *Z*?** Could it actually introduce problems?
- you are thinking you should control for *U*, **but you cannot observe it.** What to do?

Why causation matters for your thesis

If you are here, my hunch is…

- you are writing an empirical research paper or thesis
- you are interested in the effect of *X* on *Y*
- you use observational data, e.g., survey data or time-series cross-sectional data, and have lots of information about your cases
- you are asking: **should I control for *Z*?** Could it actually introduce problems?
- you are thinking you should control for *U*, **but you cannot observe it.** What to do?
- you opened social media and saw **yet another new estimator.** What to do?

Why causation matters for your thesis

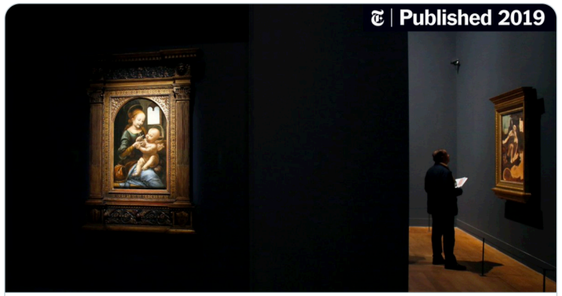
If you are here, my hunch is…

- you are writing an empirical research paper or thesis
- you are interested in the effect of *X* on *Y*
- you use observational data, e.g., survey data or time-series cross-sectional data, and have lots of information about your cases
- you are asking: **should I control for *Z*?** Could it actually introduce problems?
- you are thinking you should control for *U*, **but you cannot observe it.** What to do?
- you opened social media and saw **yet another new estimator.** What to do?
- **ultimately, you think some X might affect Y and want to find out**

Does going to the opera make you live longer?

NYT Health
@NYTHealth

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.



Another Benefit to Going to Museums? You May Live Longer (Published 2019)
Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.
nytimes.com

3:19 PM · Dec 22, 2019 · SocialFlow

318 Retweets **1,705** Quote Tweets **1,239** Likes

Does going to the opera make you live longer?



Researchers in London who followed thousands of people 50 and older over a 14-year period discovered that those who went to a museum or attended a concert just once or twice a year were 14 percent less likely to die during that period than those who didn't.

The chances of living longer only went up the more frequently people engaged with the arts, according to the study, which was published this month in The BMJ, formerly The British Medical Journal. People who went to a museum or the theater once a month or even every few months had a 31 percent reduced risk of dying in that period, according to the study.

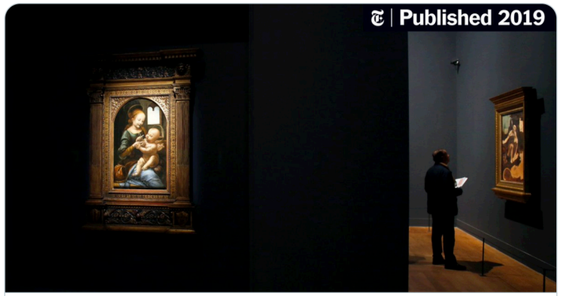
The study controlled for socioeconomic factors like a participant's income, education level and mobility, said Andrew Steptoe, a co-author of the study and the head of University College London's research department of behavioral science and health.

Source: Cramer, NYTimes (2019)

Does going to the opera make you live longer?

NYT Health
@NYTHealth

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.



Another Benefit to Going to Museums? You May Live Longer (Published 2019)
Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.
nytimes.com

3:19 PM · Dec 22, 2019 · SocialFlow

318 Retweets **1,705** Quote Tweets **1,239** Likes

RESEARCH

OPEN ACCESS
Check for updates

The art of life and death: 14 year follow-up analyses of associations between arts engagement and mortality in the English Longitudinal Study of Ageing

Daisy Fancourt,[1] Andrew Steptoe[1]

[1]Department of Behavioural Science and Health, University College London, London WC1E 7HB, UK

Correspondence to: D Fancourt
d.fancourt@ucl.ac.uk
(or @Daisy_Fancourt on Twitter; ORCID 0000-0002-6952-334X)

Cite this as: *BMJ* 2019;367:l6377
http://dx.doi.org/10.1136/bmj.l6377

Accepted: 24 September 2019

ABSTRACT

OBJECTIVE
To explore associations between different frequencies of arts engagement and mortality over a 14 year follow-up period.

DESIGN
Prospective cohort study.

PARTICIPANTS
English Longitudinal Study of Ageing cohort of 6710 community dwelling adults aged 50 years and older (53.6% women, average age 65.9 years, standard deviation 9.4) who provided baseline data in 2004-05.

INTERVENTION
Self reported receptive arts engagement (going to museums, art galleries, exhibitions, the theatre, concerts, or the opera).

MEASUREMENT
Mortality measured through data linkage to the National Health Service central register.

RESULTS
People who engaged with receptive arts activities on an infrequent basis (once or twice a year) had a 14% lower risk of dying at any point during the follow-up (809/3042 deaths, hazard ratio 0.86, 95% confidence interval 0.77 to 0.96) compared with those who never engaged (837/1762 deaths). People who engaged with receptive arts activities on a frequent basis (every few months or more) had a 31% lower risk of dying (355/1906 deaths, 0.69, 0.59 to 0.80), independent of demographic, socioeconomic, health related, behavioural, and social factors. Results were robust to a range of sensitivity analyses with no evidence of moderation by sex, socioeconomic status, or social factors. This study was observational and so causality cannot be assumed.

CONCLUSIONS
Receptive arts engagement could have a protective association with longevity in older adults. This association might be partly explained by differences in cognition, mental health, and physical activity among those who do and do not engage in the arts, but remains even when the model is adjusted for these factors.

Introduction

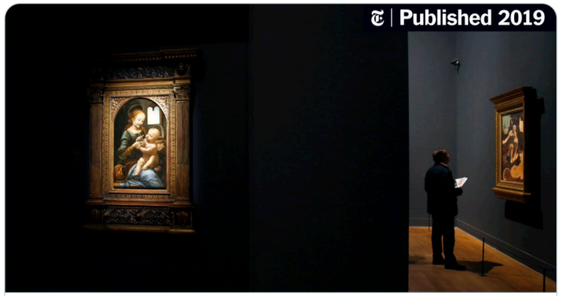
Interest in the salutogenic (health promoting) benefits of the arts is increasing. Arts activities are classified as "multimodal" health interventions because they combine multiple psychological, physical, social, and behavioural factors with an intrinsic aesthetic motivation to engage.[1] While previous studies have shown the association between arts engagement and the prevention and treatment of mental and physical health conditions, including depression, dementia, chronic pain, and frailty,[2-4] whether arts engagement actually confers survival benefits remains unclear. Some research has proposed that the universality of art and the strong emotional responses it induces are indications of its association with evolutionary

Source: Fancourt, Steptoe / BMJ (2019)

Does going to the opera make you live longer?

NYT Health
@NYTHealth

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.



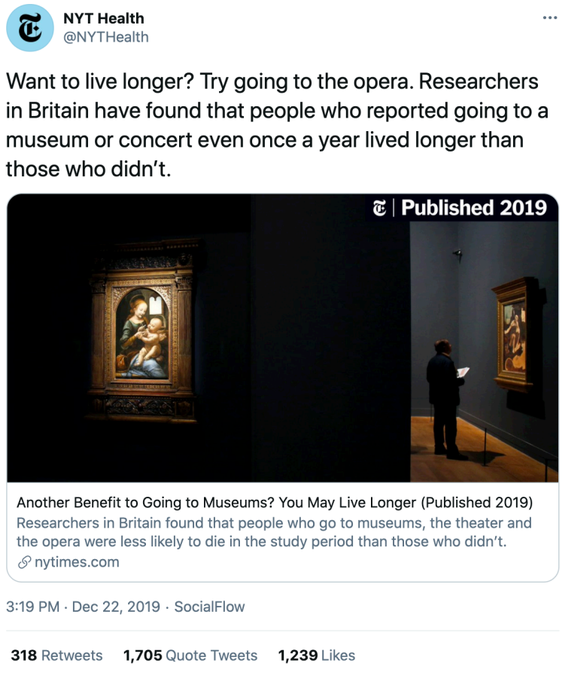
Another Benefit to Going to Museums? You May Live Longer (Published 2019)
Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.
nytimes.com

3:19 PM · Dec 22, 2019 · SocialFlow

318 Retweets **1,705** Quote Tweets **1,239** Likes



Does going to the opera make you live longer?



Different sources of association between X and Y

Does going to the opera make you live longer?

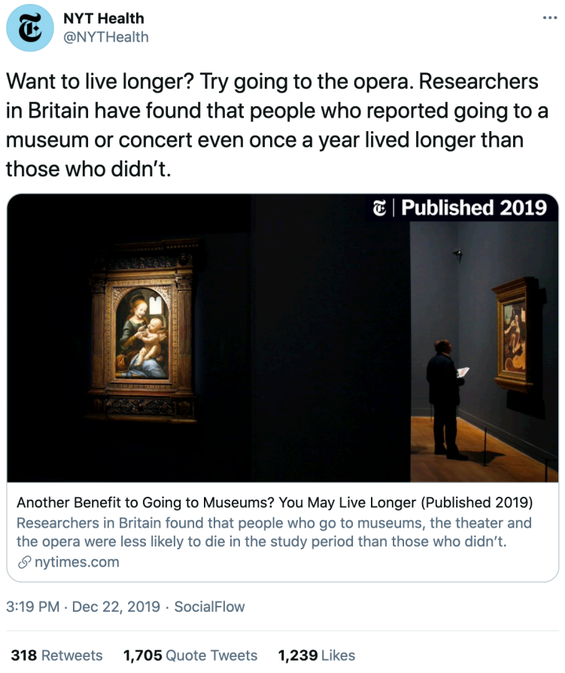


Different sources of association between X and Y

1. X may have caused Y.



Does going to the opera make you live longer?

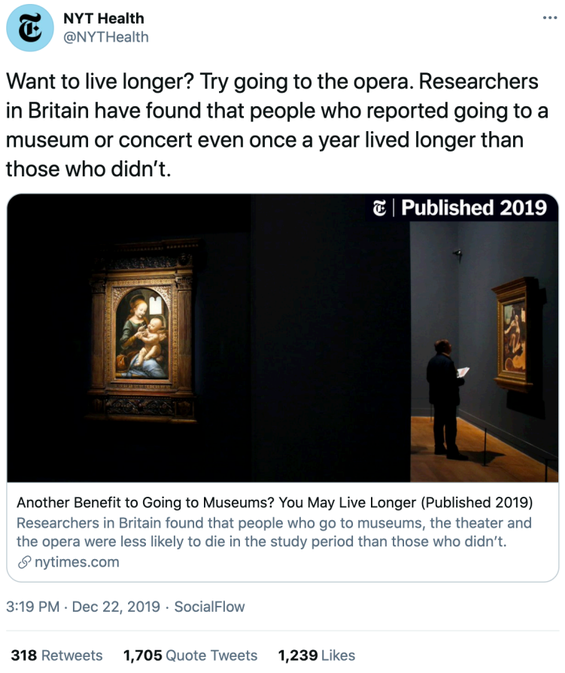


Different sources of association between X and Y

1. X may have caused Y.
2. Y may have caused X.

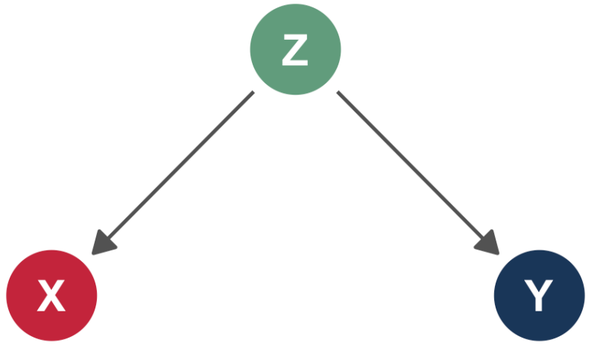


Does going to the opera make you live longer?



Different sources of association between X and Y

1. X may have caused Y.
2. Y may have caused X.
3. X and Y may share one or more common causes.

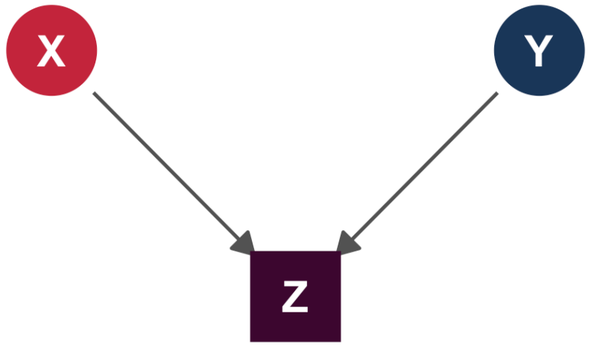


Does going to the opera make you live longer?

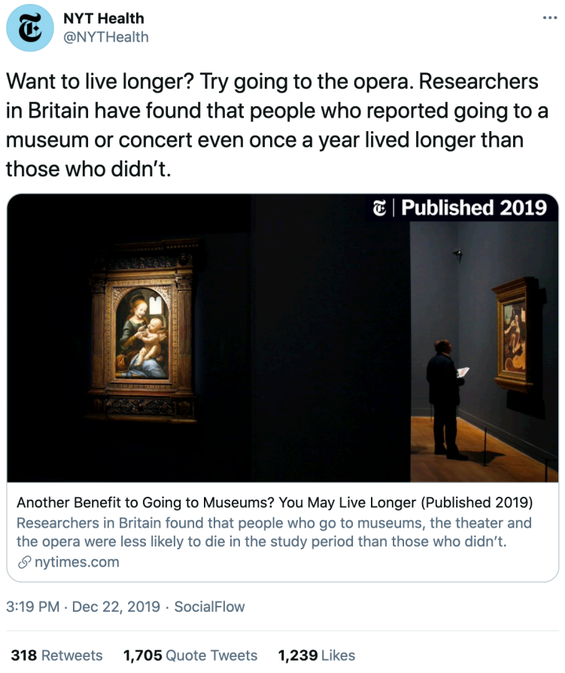


Different sources of association between X and Y

1. X may have caused Y.
2. Y may have caused X.
3. X and Y may share one or more common causes.
4. The association may have been induced by conditioning on a common effect of X and Y.



Does going to the opera make you live longer?



Different sources of association between X and Y

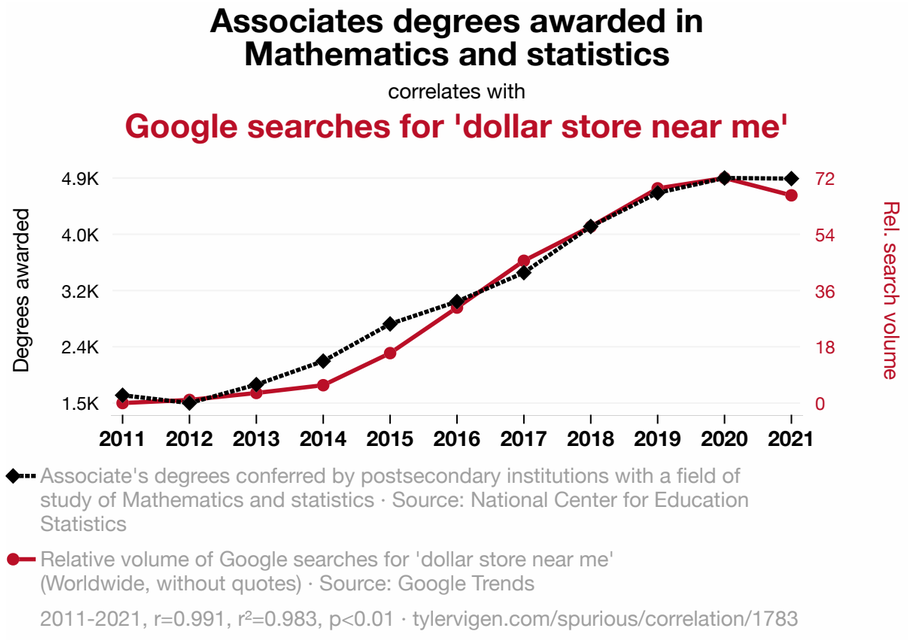
1. X may have caused Y.
2. Y may have caused X.
3. X and Y may share one or more common causes.
4. The association may have been induced by conditioning on a common effect of X and Y.
5. Chance.

Empirical regularities, causality, and the enterprise of writing papers… (?)



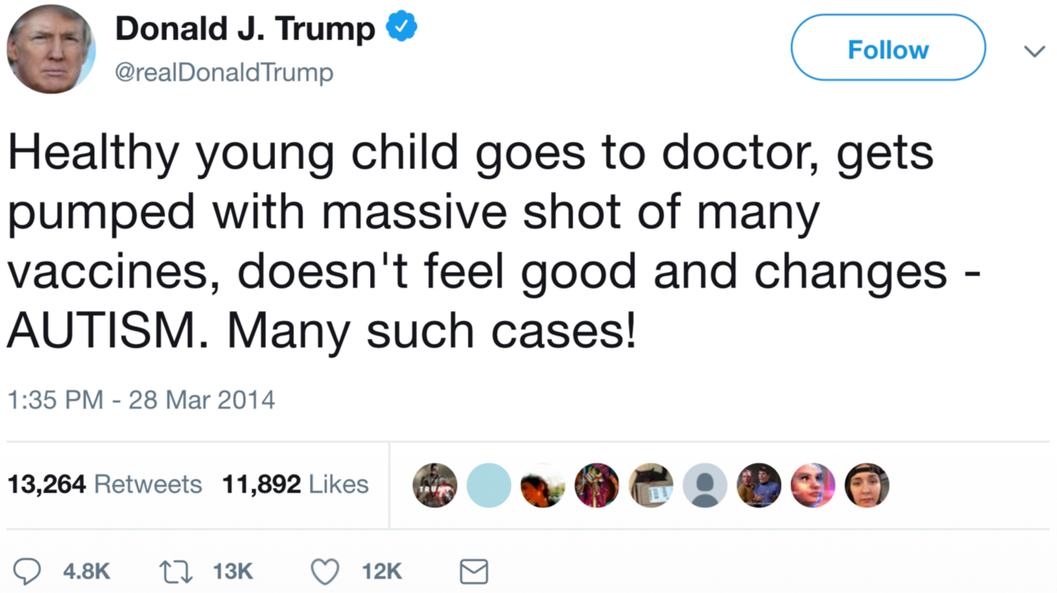
Source: Tyler Vigen | AI paper

Empirical regularities, causality, and the enterprise of writing papers… (?)



Source: Tyler Vigen | AI paper

Why causality is important



From associations to causes

From associations to causes

So, how do we figure out correlation?

From associations to causes

So, how do we figure out correlation?

Math and statistics.

From associations to causes

So, how do we figure out correlation?

Math and statistics.

And how do we figure out causation?

From associations to causes

So, how do we figure out correlation?

Math and statistics.

And how do we figure out causation?

Philosophy. Imagination. Theory. Counterfactual thinking. No (*some*) math.

From associations to causes

Causal inference

From associations to causes

Causal inference is the process of...

From associations to causes

Causal inference is the process of...

- ... ruling out non-causal sources of association,

From associations to causes

Causal inference is the process of...

- ... ruling out non-causal sources of association,
- ... highlighting the assumptions necessary to do so, and

From associations to causes

Causal inference is the process of...

- ... ruling out non-causal sources of association,
- ... highlighting the assumptions necessary to do so, and
- ... providing empirical evidence to support or refute these assumptions.

From associations to causes

Causal inference is the process of...

- ... ruling out non-causal sources of association,
- ... highlighting the assumptions necessary to do so, and
- ... providing empirical evidence to support or refute these assumptions.

- Today we will mainly think about **causal**, not statistical inference.

From associations to causes

Causal inference is the process of...

- ... ruling out non-causal sources of association,
- ... highlighting the assumptions necessary to do so, and
- ... providing empirical evidence to support or refute these assumptions.

- Today we will mainly think about **causal**, not statistical inference.
- Our understanding of causality is **probabilistic** rather than deterministic: X is a cause of Y if X is a cause of Y for some subjects in a population (or if changes in X lead to changes in the probability distribution in Y)

From associations to causes

Causal inference is the process of...

- ... ruling out non-causal sources of association,
- ... highlighting the assumptions necessary to do so, and
- ... providing empirical evidence to support or refute these assumptions.

- Today we will mainly think about **causal**, not statistical inference.
- Our understanding of causality is **probabilistic** rather than deterministic: X is a cause of Y if X is a cause of Y for some subjects in a population (or if changes in X lead to changes in the probability distribution in Y)
- Most of the time, we are interested in the **existence and magnitude of a causal effect.** Interest in the **mechanism** will come later (for now)

Counterfactual causality

Thinking in terms of counterfactuals

The Neyman-Rubin causal model

- It is a formal approach to do causal inference that fueled the **potential outcomes framework (POF)**
- Up until today, it's the mainstream framework to (statistically) talk about causation







Thinking in terms of counterfactuals

The Neyman-Rubin causal model

- It is a formal approach to do causal inference that fueled the **potential outcomes framework (POF)**
- Up until today, it's the mainstream framework to (statistically) talk about causation







The idea of potential outcomes

Thinking in terms of counterfactuals

The Neyman-Rubin causal model

- It is a formal approach to do causal inference that fueled the **potential outcomes framework (POF)**
- Up until today, it's the mainstream framework to (statistically) talk about causation







The idea of potential outcomes

- The POF assumes that each subject has a potential outcome under each treatment state

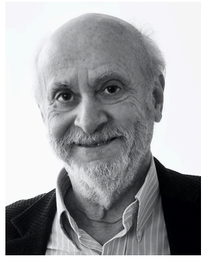
Thinking in terms of counterfactuals

The Neyman-Rubin causal model

- It is a formal approach to do causal inference that fueled the **potential outcomes framework (POF)**
- Up until today, it's the mainstream framework to (statistically) talk about causation







The idea of potential outcomes

- The POF assumes that each subject has a potential outcome under each treatment state
- Evidently, **each subject can only be observed in one treatment state at any point in time**

Thinking in terms of counterfactuals

The Neyman-Rubin causal model

- It is a formal approach to do causal inference that fueled the **potential outcomes framework (POF)**
- Up until today, it's the mainstream framework to (statistically) talk about causation

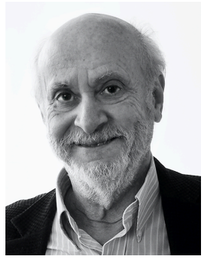


The idea of potential outcomes

- The POF assumes that each subject has a potential outcome under each treatment state
- Evidently, **each subject can only be observed in one treatment state at any point in time**



- The idea is to think about the counterfactual outcome that would have been observed had the treatment been different



Running example: roommates and the contact hypothesis

- **Context:** Prejudice between ethnic groups is persistent and politically consequential. Allport's **contact hypothesis** holds that, under the right conditions, interpersonal contact with members of an out-group reduces prejudice.
- **Intervention:** First-year students share dorm rooms. Some live with a roommate from a different ethnic group (an **out-group roommate**), others with someone from their own group.

Running example: roommates and the contact hypothesis

- **Context:** Prejudice between ethnic groups is persistent and politically consequential. Allport's **contact hypothesis** holds that, under the right conditions, interpersonal contact with members of an out-group reduces prejudice.
- **Intervention:** First-year students share dorm rooms. Some live with a roommate from a different ethnic group (an **out-group roommate**), others with someone from their own group.

Research question: Does living with an out-group roommate reduce a student's **prejudice**?

Defining causal effects: individual treatment effects

Defining causal effects: individual treatment effects

- Assume that each subject i in a population of interest can be exposed to two alternative states of a cause: **treatment**, $D_i = 1$ (out-group roommate), and **control**, $D_i = 0$ (in-group roommate)

Defining causal effects: individual treatment effects

- Assume that each subject i in a population of interest can be exposed to two alternative states of a cause: **treatment**, $D_i = 1$ (out-group roommate), and **control**, $D_i = 0$ (in-group roommate)
- The POF assumes that each subject has a potential outcome under both treatment states, $Y_i(1)$ and $Y_i(0)$

Defining causal effects: individual treatment effects

- Assume that each subject i in a population of interest can be exposed to two alternative states of a cause: **treatment**, $D_i = 1$ (out-group roommate), and **control**, $D_i = 0$ (in-group roommate)
- The POF assumes that each subject has a potential outcome under both treatment states, $Y_i(1)$ and $Y_i(0)$
- An **individual treatment effect (ITE)** can thus be defined as $\tau_i = Y_i(1) - Y_i(0)$

Defining causal effects: individual treatment effects

- Assume that each subject i in a population of interest can be exposed to two alternative states of a cause: **treatment**, $D_i = 1$ (out-group roommate), and **control**, $D_i = 0$ (in-group roommate)
- The POF assumes that each subject has a potential outcome under both treatment states, $Y_i(1)$ and $Y_i(0)$
- An **individual treatment effect (ITE)** can thus be defined as $\tau_i = Y_i(1) - Y_i(0)$
- Evidently, each subject can only be observed in one treatment state *at any point in time*

Defining causal effects: individual treatment effects

- Assume that each subject i in a population of interest can be exposed to two alternative states of a cause: **treatment**, $D_i = 1$ (out-group roommate), and **control**, $D_i = 0$ (in-group roommate)
- The POF assumes that each subject has a potential outcome under both treatment states, $Y_i(1)$ and $Y_i(0)$
- An **individual treatment effect (ITE)** can thus be defined as $\tau_i = Y_i(1) - Y_i(0)$
- Evidently, each subject can only be observed in one treatment state *at any point in time*
- Therefore, the ITE is a non-observable counterfactual (what Holland, 1986, labels the **fundamental problem of causal inference**)

Defining causal effects: individual treatment effects

- Assume that each subject i in a population of interest can be exposed to two alternative states of a cause: **treatment**, $D_i = 1$ (out-group roommate), and **control**, $D_i = 0$ (in-group roommate)
- The POF assumes that each subject has a potential outcome under both treatment states, $Y_i(1)$ and $Y_i(0)$
- An **individual treatment effect (ITE)** can thus be defined as $\tau_i = Y_i(1) - Y_i(0)$
- Evidently, each subject can only be observed in one treatment state *at any point in time*
- Therefore, the ITE is a non-observable counterfactual (what Holland, 1986, labels the **fundamental problem of causal inference**)
- Note that even successive exposure of the same subjects to alternative causal states does not allow for the observation of the ITE — **Why?**

Defining causal effects: notation

Defining causal effects: notation

- Binary causal variable, $D_i = 1$ for students with an out-group roommate, $D_i = 0$ for students with an in-group roommate

Defining causal effects: notation

- Binary causal variable, $D_i = 1$ for students with an out-group roommate, $D_i = 0$ for students with an in-group roommate
- Observable outcome variable:

Defining causal effects: notation

- Binary causal variable, $D_i = 1$ for students with an out-group roommate, $D_i = 0$ for students with an in-group roommate
- Observable outcome variable:

$$Y_i = Y_i(1) \text{ if } D_i = 1$$

Defining causal effects: notation

- Binary causal variable, $D_i = 1$ for students with an out-group roommate, $D_i = 0$ for students with an in-group roommate
- Observable outcome variable:

$$Y_i = Y_i(1) \text{ if } D_i = 1$$
$$Y_i = Y_i(0) \text{ if } D_i = 0$$

Defining causal effects: notation

- Binary causal variable, $D_i = 1$ for students with an out-group roommate, $D_i = 0$ for students with an in-group roommate
- Observable outcome variable:

$$
\begin{aligned}
Y_i &= Y_i(1) \text{ if } D_i = 1 \\
Y_i &= Y_i(0) \text{ if } D_i = 0 \\
Y_i &= D_i\, Y_i(1) + (1 - D_i)\, Y_i(0)
\end{aligned}
$$

Defining causal effects: notation

- Binary causal variable, $D_i = 1$ for students with an out-group roommate, $D_i = 0$ for students with an in-group roommate
- Observable outcome variable:

$$Y_i = Y_i(1) \text{ if } D_i = 1$$
$$Y_i = Y_i(0) \text{ if } D_i = 0$$
$$Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0)$$

Group	Under treatment $(D = 1)$	Under control $(D = 0)$
Treated $(D = 1)$	Observable as $Y_i(1)$	Counterfactual $Y_i(0)$
Untreated $(D = 0)$	Counterfactual $Y_i(1)$	Observable as $Y_i(0)$

Defining causal effects: notation

- Binary causal variable, $D_i = 1$ for students with an out-group roommate, $D_i = 0$ for students with an in-group roommate
- Observable outcome variable:

$$Y_i = Y_i(1) \text{ if } D_i = 1$$
$$Y_i = Y_i(0) \text{ if } D_i = 0$$
$$Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0)$$

Group	Prejudice with an out-group roommate $(D = 1)$	Prejudice with an in-group roommate $(D = 0)$
Has an out-group roommate $(D = 1)$	Observable as $Y_i(1)$	Counterfactual $Y_i(0)$
Has an in-group roommate $(D = 0)$	Counterfactual $Y_i(1)$	Observable as $Y_i(0)$

Defining causal effects: notation

Why is the causal effect of D on Y not simply the difference between the upper-left and lower-right cell in the table?

Group	Prejudice with an out-group roommate $(D = 1)$	Prejudice with an in-group roommate $(D = 0)$
Has an out-group roommate $(D = 1)$	Observable as $Y_i(1)$	Counterfactual $Y_i(0)$
Has an in-group roommate $(D = 0)$	Counterfactual $Y_i(1)$	Observable as $Y_i(0)$

Defining causal effects: average treatment effects

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**
- What we can immediately calculate from observed Y and D is the **naïve estimator of the ATE**:

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**
- What we can immediately calculate from observed Y and D is the **naïve estimator of the ATE**:

$$\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**
- What we can immediately calculate from observed Y and D is the **naïve estimator of the ATE**:

$$\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

- **NATE will correspond to ATE only under very specific conditions** regarding the process by which subjects are assigned to causal states. Think of:

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**
- What we can immediately calculate from observed Y and D is the **naïve estimator of the ATE**:

$$\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

- **NATE will correspond to ATE only under very specific conditions** regarding the process by which subjects are assigned to causal states. Think of:
  - A subject decides itself whether to enter one causal state or another (students choose their roommates)

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**
- What we can immediately calculate from observed Y and D is the **naïve estimator of the ATE**:

$$\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

- **NATE will correspond to ATE only under very specific conditions** regarding the process by which subjects are assigned to causal states. Think of:
  - A subject decides itself whether to enter one causal state or another (students choose their roommates)
  - An outside actor chooses to allocate subjects to one state or another (the housing office)

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**
- What we can immediately calculate from observed Y and D is the **naïve estimator of the ATE**:

$$\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

- **NATE will correspond to ATE only under very specific conditions** regarding the process by which subjects are assigned to causal states. Think of:
  - A subject decides itself whether to enter one causal state or another (students choose their roommates)
  - An outside actor chooses to allocate subjects to one state or another (the housing office)
  - A random allocation by some EUI researcher decides

Defining causal effects: average treatment effects

- In the counterfactual modeling tradition, attention is usually on estimating various **average treatment effects**, $\text{ATE} = E[Y_i(1)] - E[Y_i(0)]$, for groups of subjects
- **ATE so defined is not observable either** — **Why?**
- What we can immediately calculate from observed Y and D is the **naïve estimator of the ATE**:

$$\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

- **NATE will correspond to ATE only under very specific conditions** regarding the process by which subjects are assigned to causal states. Think of:
  - A subject decides itself whether to enter one causal state or another (students choose their roommates)
  - An outside actor chooses to allocate subjects to one state or another (the housing office)
  - A random allocation by some EUI researcher decides
  - Some combination thereof

Eight students: the world with full information

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

Eight students: the world with full information

Eight first-year students. We imagine we can see both potential outcomes (prejudice, 0–10).

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

Eight students: the world with full information

Eight first-year students. We imagine we can see both potential outcomes (prejudice, 0–10).

Students **chose** their roommates.

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

Eight students: the world with full information

Eight first-year students. We imagine we can see both potential outcomes (prejudice, 0–10).

Students **chose** their roommates.

Average of the last column:

$$\text{ATE} = \frac{-4}{8} = -0.5$$

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

…and what we actually observe

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6			0
2		2		1
3	4			0
4	6			0
5		1		1
6		2		1
7	8			0
8	4			0

…and what we actually observe

Compare the two groups as we find them:

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6			0
2		2		1
3	4			0
4	6			0
5		1		1
6		2		1
7	8			0
8	4			0

…and what we actually observe

Compare the two groups as we find them:

$$\frac{2+1+2}{3} - \frac{6+4+6+8+4}{5} = 1.67 - 5.6$$

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6			0
2		2		1
3	4			0
4	6			0
5		1		1
6		2		1
7	8			0
8	4			0

…and what we actually observe

Compare the two groups as we find them:

$$\frac{2+1+2}{3} - \frac{6+4+6+8+4}{5} = 1.67 - 5.6$$

$$\text{NATE} = -3.93$$

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6			0
2		2		1
3	4			0
4	6			0
5		1		1
6		2		1
7	8			0
8	4			0

…and what we actually observe

Compare the two groups as we find them:

$$\frac{2+1+2}{3} - \frac{6+4+6+8+4}{5} = 1.67 - 5.6$$

$$\text{NATE} = -3.93$$

The naive comparison says contact cuts prejudice by almost 4 points. The true average effect is **half a point**.

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6			0
2		2		1
3	4			0
4	6			0
5		1		1
6		2		1
7	8			0
8	4			0

ATE, ATT, ATC: three target populations

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

ATE, ATT, ATC: three target populations

- Same unit-level effects, averaged over different people:

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

ATE, ATT, ATC: three target populations

- Same unit-level effects, averaged over different people:

$$\text{ATE} = E[Y_i(1) - Y_i(0)]$$
$$\text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$
$$\text{ATC} = E[Y_i(1) - Y_i(0) \mid D_i = 0]$$

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

ATE, ATT, ATC: three target populations

- Same unit-level effects, averaged over different people:

$$\text{ATE} = E[Y_i(1) - Y_i(0)]$$
$$\text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$
$$\text{ATC} = E[Y_i(1) - Y_i(0) \mid D_i = 0]$$

- **Your turn (1 min)**: compute the ATT and the ATC from the table.

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

ATE, ATT, ATC: three target populations

- Same unit-level effects, averaged over different people:

$$\text{ATE} = E[Y_i(1) - Y_i(0)]$$
$$\text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$
$$\text{ATC} = E[Y_i(1) - Y_i(0) \mid D_i = 0]$$

- **Your turn (1 min)**: compute the ATT and the ATC from the table.
- ATT $= \frac{-2-2+0}{3} = -1.33$ · ATC $= \frac{-1+0+1-1+1}{5} = 0$

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

ATE, ATT, ATC: three target populations

- Same unit-level effects, averaged over different people:

$$\text{ATE} = E[Y_i(1) - Y_i(0)]$$
$$\text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$
$$\text{ATC} = E[Y_i(1) - Y_i(0) \mid D_i = 0]$$

- **Your turn (1 min)**: compute the ATT and the ATC from the table.
- ATT $= \frac{-2-2+0}{3} = -1.33$ · ATC $= \frac{-1+0+1-1+1}{5} = 0$
- Contact "works" for those who sought it out, and **not for the rest**. Which one is your estimand?

Student	$Y_i(0)$	$Y_i(1)$	τ_i	D_i
1	6	5	−1	0
2	4	2	−2	1
3	4	4	0	0
4	6	7	1	0
5	3	1	−2	1
6	2	2	0	1
7	8	7	−1	0
8	4	5	1	0

Defining biases

- We can define the ATE as a weighted average of ATT and ATC:

$$\begin{aligned}\text{ATE} &= \pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC} \\ &= \pi\big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + (1-\pi)\big(E[Y_i(1) \mid D_i = 0] - E[Y_i(0) \mid D_i = 0]\big)\end{aligned}$$

where π is the proportion of subjects under treatment

Defining biases

- We can define the ATE as a weighted average of ATT and ATC:

$$\begin{aligned}\text{ATE} &= \pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC} \\ &= \pi\big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + (1-\pi)\big(E[Y_i(1) \mid D_i = 0] - E[Y_i(0) \mid D_i = 0]\big)\end{aligned}$$

where π is the proportion of subjects under treatment

- In our example, $\text{ATT} = -1.33$, $\text{ATC} = 0$ and $\pi = 3/8$, so:

Defining biases

- We can define the ATE as a weighted average of ATT and ATC:

$$\begin{aligned}\text{ATE} &= \pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC} \\ &= \pi\big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + (1-\pi)\big(E[Y_i(1) \mid D_i = 0] - E[Y_i(0) \mid D_i = 0]\big)\end{aligned}$$

where π is the proportion of subjects under treatment

- In our example, $\text{ATT} = -1.33$, $\text{ATC} = 0$ and $\pi = 3/8$, so:

$$\text{ATE} = \tfrac{3}{8} \cdot (-1.33) + \tfrac{5}{8} \cdot 0 = -0.5$$

Defining biases

- We can define the ATE as a weighted average of ATT and ATC:

$$\begin{aligned}\text{ATE} &= \pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC} \\ &= \pi\big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + (1-\pi)\big(E[Y_i(1) \mid D_i = 0] - E[Y_i(0) \mid D_i = 0]\big)\end{aligned}$$

where π is the proportion of subjects under treatment

- In our example, $\text{ATT} = -1.33$, $\text{ATC} = 0$ and $\pi = 3/8$, so:

$$\text{ATE} = \tfrac{3}{8} \cdot (-1.33) + \tfrac{5}{8} \cdot 0 = -0.5$$

- Recall that $\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] = 1.67 - 5.6 = -3.93$

Defining biases

- We can define the ATE as a weighted average of ATT and ATC:

$$\begin{aligned}\text{ATE} &= \pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC} \\ &= \pi\big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + (1-\pi)\big(E[Y_i(1) \mid D_i = 0] - E[Y_i(0) \mid D_i = 0]\big)\end{aligned}$$

where π is the proportion of subjects under treatment

- In our example, $\text{ATT} = -1.33$, $\text{ATC} = 0$ and $\pi = 3/8$, so:

$$\text{ATE} = \tfrac{3}{8} \cdot (-1.33) + \tfrac{5}{8} \cdot 0 = -0.5$$

- Recall that $\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] = 1.67 - 5.6 = -3.93$
- **Total bias** of NATE as an estimate of ATE is simply $\text{NATE} - \text{ATE} = -3.93 - (-0.5) = -3.43$

Defining biases

- We can define the ATE as a weighted average of ATT and ATC:

$$\begin{aligned}\text{ATE} &= \pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC} \\ &= \pi\big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + (1-\pi)\big(E[Y_i(1) \mid D_i = 0] - E[Y_i(0) \mid D_i = 0]\big)\end{aligned}$$

where π is the proportion of subjects under treatment

- In our example, $\text{ATT} = -1.33$, $\text{ATC} = 0$ and $\pi = 3/8$, so:

$$\text{ATE} = \tfrac{3}{8} \cdot (-1.33) + \tfrac{5}{8} \cdot 0 = -0.5$$

- Recall that $\text{NATE} = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] = 1.67 - 5.6 = -3.93$
- **Total bias** of NATE as an estimate of ATE is simply $\text{NATE} - \text{ATE} = -3.93 - (-0.5) = -3.43$
- Using the above expressions, we can further **decompose total bias into its component parts**

Defining biases

$$
\begin{aligned}
\text{Total bias} &= \text{NATE} - \text{ATE} \\
&= E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] - \{\pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC}\} && \text{definitions} \\
&= \big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\cdots\} && \pm E[Y_i(0) \mid D_i = 1] \\
&= \text{ATT} + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\text{ATT} - (1-\pi)(\text{ATT} - \text{ATC})\} && \pi = 1 - (1-\pi) \\
&= \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline bias } = 3 - 5.6 \,=\, -2.6} + \underbrace{(1-\pi)\big(\text{ATT} - \text{ATC}\big)}_{\text{differential treatment effect bias } = \frac{5}{8}(-1.33) \,=\, -0.83} && \text{ATTs cancel}
\end{aligned}
$$

Defining biases

$$
\begin{aligned}
\text{Total bias} &= \text{NATE} - \text{ATE} \\
&= E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] - \{\pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC}\} && \text{definitions} \\
&= \big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\cdots\} && \pm E[Y_i(0) \mid D_i = 1] \\
&= \text{ATT} + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\text{ATT} - (1-\pi)(\text{ATT} - \text{ATC})\} && \pi = 1 - (1-\pi) \\
&= \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline bias } = 3 - 5.6 \,=\, -2.6} + \underbrace{(1-\pi)\big(\text{ATT} - \text{ATC}\big)}_{\text{differential treatment effect bias } = \frac{5}{8}(-1.33) \,=\, -0.83} && \text{ATTs cancel}
\end{aligned}
$$

- where **baseline bias** is the difference in the average outcome *in the absence of treatment* between those under treatment and those under control: students who chose out-group roommates were **less prejudiced to begin with**
- and **differential treatment effect bias** is the difference in the average treatment effect between treated and control groups, weighted by the share in the control state: contact **helps more** those who sought it
- $\therefore$ Total bias = **baseline bias** + **differential treatment effect bias** $= -2.6 - 0.83 = -3.43$

Defining biases

$$
\begin{aligned}
\text{Total bias} &= \text{NATE} - \text{ATE} \\
&= E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] - \{\pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC}\} && \text{definitions} \\
&= \big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\cdots\} && \pm E[Y_i(0) \mid D_i = 1] \\
&= \text{ATT} + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\text{ATT} - (1-\pi)(\text{ATT} - \text{ATC})\} && \pi = 1 - (1-\pi) \\
&= \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline bias } = 3 - 5.6 \,=\, -2.6} + \underbrace{(1-\pi)\big(\text{ATT} - \text{ATC}\big)}_{\text{differential treatment effect bias } = \frac{5}{8}(-1.33) \,=\, -0.83} && \text{ATTs cancel}
\end{aligned}
$$

What happens when “total bias” is equal to zero?

Defining biases

$$
\begin{aligned}
\text{Total bias} &= \text{NATE} - \text{ATE} \\
&= E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] - \{\pi \cdot \text{ATT} + (1-\pi) \cdot \text{ATC}\} && \text{definitions} \\
&= \big(E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]\big) + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\cdots\} && \pm E[Y_i(0) \mid D_i = 1] \\
&= \text{ATT} + \big(E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]\big) - \{\text{ATT} - (1-\pi)(\text{ATT} - \text{ATC})\} && \pi = 1 - (1-\pi) \\
&= \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{baseline bias } = 3 - 5.6 \,=\, -2.6} + \underbrace{(1-\pi)\big(\text{ATT} - \text{ATC}\big)}_{\text{differential treatment effect bias } = \frac{5}{8}(-1.33) \,=\, -0.83} && \text{ATTs cancel}
\end{aligned}
$$

What happens when “total bias” is equal to zero?

$$\text{NATE} = \text{ATE}$$

What we can estimate with “realized” outcomes = True effect

SUTVA: when is $Y_i(d)$ even well-defined?

The **stable unit treatment value assumption** has two parts:

SUTVA: when is $Y_i(d)$ even well-defined?

The **stable unit treatment value assumption** has two parts:

- **No interference**: student i's prejudice does not depend on who *other* students room with.

SUTVA: when is $Y_i(d)$ even well-defined?

The **stable unit treatment value assumption** has two parts:

- **No interference**: student i's prejudice does not depend on who *other* students room with.
- **One version of the treatment**: "out-group roommate" means the same thing for everyone.

SUTVA: when is $Y_i(d)$ even well-defined?

The **stable unit treatment value assumption** has two parts:

- **No interference**: student i's prejudice does not depend on who *other* students room with.
- **One version of the treatment**: "out-group roommate" means the same thing for everyone.

> In dorms, both are doubtful: friends of friends, shared corridors, and roommates who barely talk vs. become close friends.

SUTVA: when is $Y_i(d)$ even well-defined?

The **stable unit treatment value assumption** has two parts:

- **No interference**: student i's prejudice does not depend on who *other* students room with.
- **One version of the treatment**: "out-group roommate" means the same thing for everyone.

> In dorms, both are doubtful: friends of friends, shared corridors, and roommates who barely talk vs. become close friends.

If SUTVA fails, the **estimand itself** is not defined, before any data problem.

Ignorability: what randomization buys

Ignorability: what randomization buys

- If $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$, both biases are zero and $\mathbf{NATE} = \mathbf{ATE}$.

Ignorability: what randomization buys

- If $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$, both biases are zero and $\mathbf{NATE} = \mathbf{ATE}$.
- **Randomization** guarantees this by design: a coin, not the student, decides D_i.

Ignorability: what randomization buys

- If $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$, both biases are zero and $\mathbf{NATE} = \mathbf{ATE}$.
- **Randomization** guarantees this by design: a coin, not the student, decides D_i.
- Without randomization we hope for **conditional ignorability**:

Ignorability: what randomization buys

- If $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$, both biases are zero and $\mathbf{NATE} = \mathbf{ATE}$.
- **Randomization** guarantees this by design: a coin, not the student, decides D_i.
- Without randomization we hope for **conditional ignorability**:

$$\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i \mid X_i$$

within groups of students with the same X, roommate type is as good as random

Ignorability: what randomization buys

- If $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$, both biases are zero and $\mathbf{NATE} = \mathbf{ATE}$.
- **Randomization** guarantees this by design: a coin, not the student, decides D_i.
- Without randomization we hope for **conditional ignorability**:

$$\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i \mid X_i$$

within groups of students with the same X, roommate type is as good as random

But **which X?** That is what DAGs answer.

What is your estimand?

A coefficient is an answer. What was the question?

A coefficient is an answer. What was the question?

Running example: does living with an **out-group roommate** reduce **prejudice**? (Allport's contact hypothesis)

A coefficient is an answer. What was the question?

Running example: does living with an **out-group roommate** reduce **prejudice**? (Allport's contact hypothesis)

```
lm(prejudice ~ outgroup_roommate + openness + outgroup_friends, data = dorms)
```

A coefficient is an answer. What was the question?

Running example: does living with an **out-group roommate** reduce **prejudice**? (Allport's contact hypothesis)

```
lm(prejudice ~ outgroup_roommate + openness + outgroup_friends, data = dorms)
```

Description

How much less prejudiced are students with out-group roommates than similar students without?

A coefficient is an answer. What was the question?

Running example: does living with an **out-group roommate** reduce **prejudice**? (Allport's contact hypothesis)

```
lm(prejudice ~ outgroup_roommate + openness + outgroup_friends, data = dorms)
```

Description

How much less prejudiced are students with out-group roommates than similar students without?

Prediction

Does knowing a student's roommate help us forecast their prejudice score?

A coefficient is an answer. What was the question?

Running example: does living with an **out-group roommate** reduce **prejudice**? (Allport's contact hypothesis)

```
lm(prejudice ~ outgroup_roommate + openness + outgroup_friends, data = dorms)
```

Description

How much less prejudiced are students with out-group roommates than similar students without?

Prediction

Does knowing a student's roommate help us forecast their prejudice score?

Causal effect

How much would a student's prejudice change if we assigned them an out-group roommate?

A coefficient is an answer. What was the question?

Running example: does living with an **out-group roommate** reduce **prejudice**? (Allport's contact hypothesis)

```
lm(prejudice ~ outgroup_roommate + openness + outgroup_friends, data = dorms)
```

Description

How much less prejudiced are students with out-group roommates than similar students without?

Prediction

Does knowing a student's roommate help us forecast their prejudice score?

Causal effect

How much would a student's prejudice change if we assigned them an out-group roommate?

Same (*code*) test, three questions. **The (*code*) test does not tell you which one you answered.**

$\hat{\beta}$ is not an estimand.
It is an estimate **of something**.

$\hat{\beta}$ is not an estimand.
It is an estimate **of something**.

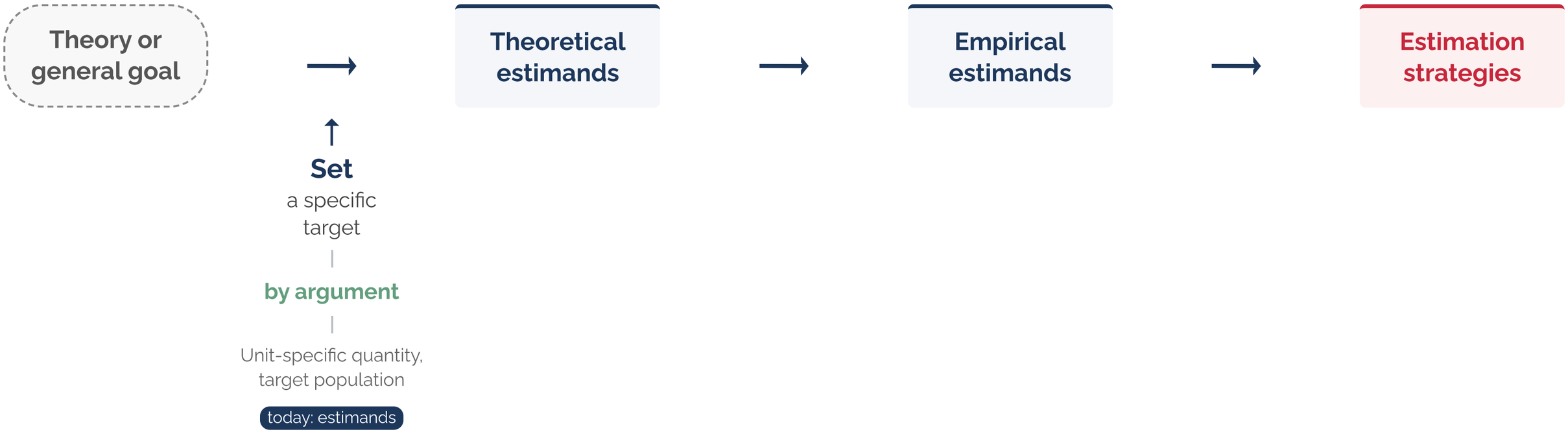
Say what.

Set, link, learn: from theory to evidence

Theory or general goal → Theoretical estimands → Empirical estimands → Estimation strategies

Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

Set, link, learn: from theory to evidence



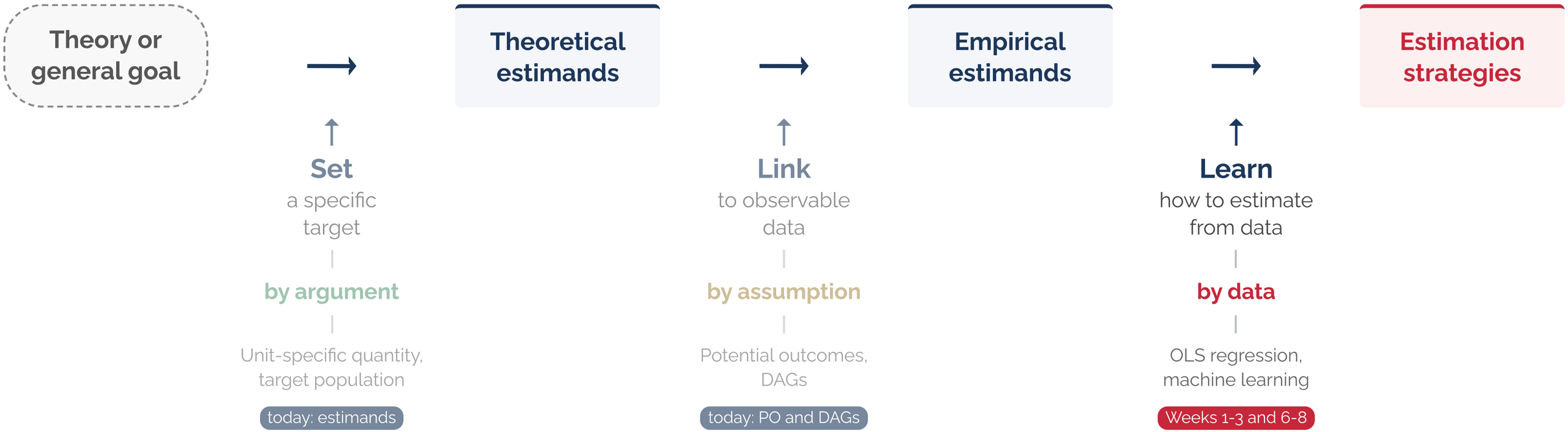
Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

Set, link, learn: from theory to evidence



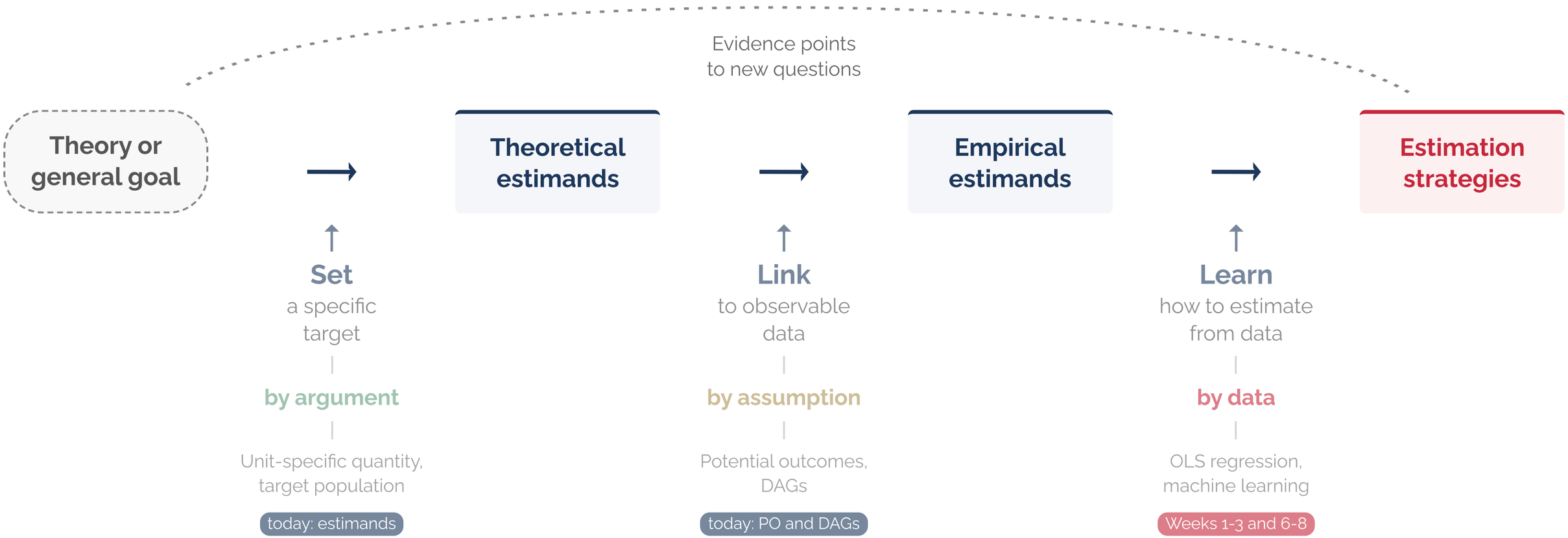
Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

Set, link, learn: from theory to evidence



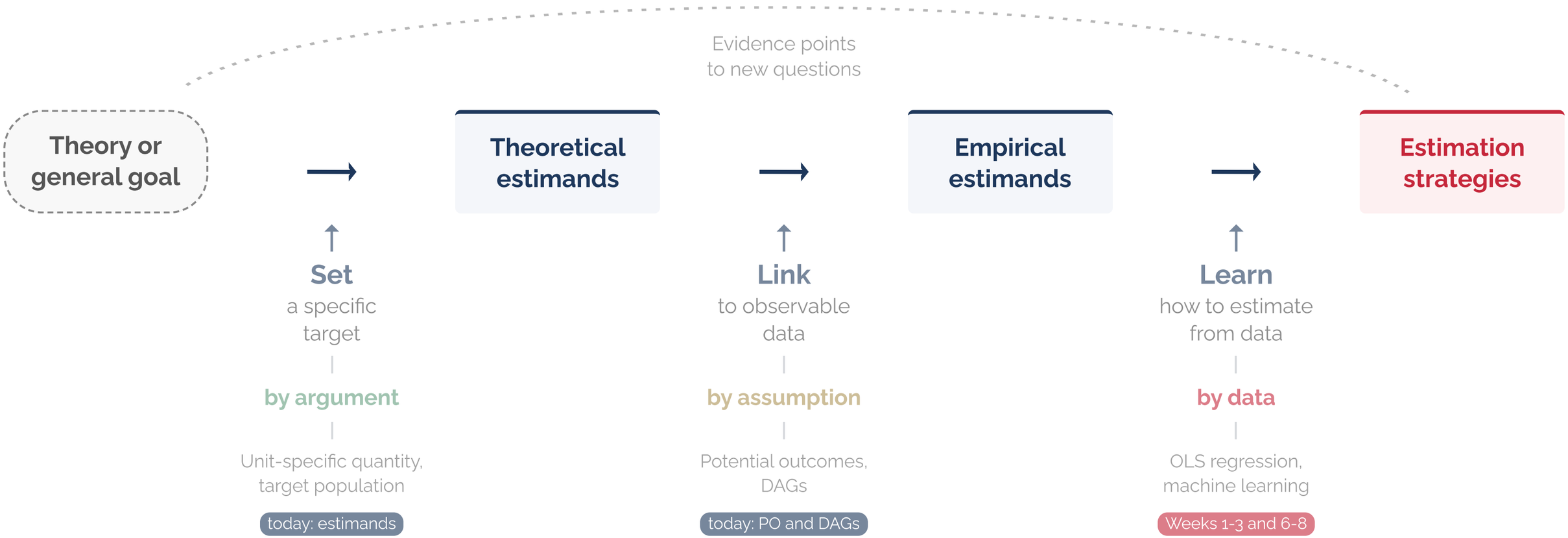
Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

Set, link, learn: from theory to evidence



Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

Set, link, learn: from theory to evidence



Most papers start with the estimation strategy. **Lundberg, Johnson & Stewart** ask us to start by **setting** the target.

Lundberg, Johnson & Stewart (2021), "What is your estimand?", *American Sociological Review* 86(3)

A theoretical estimand has two parts

A theoretical estimand has two parts

1. Unit-specific quantity

What we would want to know about each unit i, e.g. how student i's prejudice would differ with an out-group roommate.

A theoretical estimand has two parts

1. Unit-specific quantity

What we would want to know about each unit i, e.g. how student i's prejudice would differ with an out-group roommate.

2. Target population $\mathcal{P}$

Over whom we aggregate, e.g. all first-year students in university dorms across the EU (?).

A theoretical estimand has two parts

1. Unit-specific quantity

What we would want to know about each unit i, e.g. how student i's prejudice would differ with an out-group roommate.

2. Target population $\mathcal{P}$

Over whom we aggregate, e.g. all first-year students in university dorms across the EU (?).

$$\tau = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \Big[Y_i(1) - Y_i(0) \Big]$$

where $Y_i(1)$ and $Y_i(0)$ are student i's prejudice with and without an out-group roommate

A theoretical estimand has two parts

1. Unit-specific quantity

What we would want to know about each unit i, e.g. how student i's prejudice would differ with an out-group roommate.

2. Target population $\mathcal{P}$

Over whom we aggregate, e.g. all first-year students in university dorms across the EU (?).

$$\tau = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \Big[Y_i(1) - Y_i(0) \Big]$$

where $Y_i(1)$ and $Y_i(0)$ are student i's prejudice with and without an out-group roommate

Notice what is missing: **no regression, no β, no control variables.**

Descriptive estimands are estimands too

Descriptive estimands are estimands too

Not every good question is causal. A **descriptive** estimand:

Descriptive estimands are estimands too

Not every good question is causal. A **descriptive** estimand:

$$\tau_{\text{gap}} \;=\; E\big[Y_i \mid \text{out-group roommate}\big] - E\big[Y_i \mid \text{in-group roommate}\big], \quad i \in \mathcal{P}$$

the prejudice gap between the two groups of students, as they are

Descriptive estimands are estimands too

Not every good question is causal. A **descriptive** estimand:

$$\tau_{\text{gap}} \;=\; E\big[Y_i \mid \text{out-group roommate}\big] - E\big[Y_i \mid \text{in-group roommate}\big], \quad i \in \mathcal{P}$$

the prejudice gap between the two groups of students, as they are

- No counterfactual, no DAG, no “controls” needed.

Descriptive estimands are estimands too

Not every good question is causal. A **descriptive** estimand:

$$\tau_{\text{gap}} \;=\; E\big[Y_i \mid \text{out-group roommate}\big] - E\big[Y_i \mid \text{in-group roommate}\big], \quad i \in \mathcal{P}$$

the prejudice gap between the two groups of students, as they are

- No counterfactual, no DAG, no "controls" needed.
- Perfectly legitimate, if you **say so** and do not slip into causal language.

Descriptive estimands are estimands too

Not every good question is causal. A **descriptive** estimand:

$$\tau_{\text{gap}} \;=\; E\big[Y_i \mid \text{out-group roommate}\big] - E\big[Y_i \mid \text{in-group roommate}\big], \quad i \in \mathcal{P}$$

the prejudice gap between the two groups of students, as they are

- No counterfactual, no DAG, no "controls" needed.
- Perfectly legitimate, if you **say so** and do not slip into causal language.
- The trouble starts when a descriptive estimate is **read as a causal one**.

The target population is part of the question

Boisjoly, Duncan, Kremer, Levy & Eccles (2006), *AER* 96(5) · Behaghel, Crépon & Le Barbanchon (2015), *AEJ: Applied* 7(3)

The target population is part of the question

Roommates

Boisjoly et al. (2006) use roommates who were **randomly assigned**, but only among students whose stated housing preferences allowed it.

→ The estimand is the effect **for those students**, not for everyone.

Boisjoly, Duncan, Kremer, Levy & Eccles (2006), *AER* 96(5) · Behaghel, Crépon & Le Barbanchon (2015), *AEJ: Applied* 7(3)

The target population is part of the question

Roommates

Boisjoly et al. (2006) use roommates who were **randomly assigned**, but only among students whose stated housing preferences allowed it.

→ The estimand is the effect **for those students**, not for everyone.

Anonymous CVs

Behaghel, Crépon & Le Barbanchon (2015) randomize anonymous CVs among firms that **volunteered** for the French program.

→ The estimand is the effect **among volunteers**, not the effect of a mandate for all firms.

Boisjoly, Duncan, Kremer, Levy & Eccles (2006), *AER* 96(5) · Behaghel, Crépon & Le Barbanchon (2015), *AEJ: Applied* 7(3)

The target population is part of the question

Roommates

Boisjoly et al. (2006) use roommates who were **randomly assigned**, but only among students whose stated housing preferences allowed it.

→ The estimand is the effect **for those students**, not for everyone.

Anonymous CVs

Behaghel, Crépon & Le Barbanchon (2015) randomize anonymous CVs among firms that **volunteered** for the French program.

→ The estimand is the effect **among volunteers**, not the effect of a mandate for all firms.

A perfectly identified effect can still answer **the wrong population's question**. We come back to this as **transportability** in Week 4.

Boisjoly, Duncan, Kremer, Levy & Eccles (2006), *AER* 96(5) · Behaghel, Crépon & Le Barbanchon (2015), *AEJ: Applied* 7(3)

From theoretical to empirical estimand

From theoretical to empirical estimand

The **empirical estimand** is written only in terms of things we can observe:

From theoretical to empirical estimand

The **empirical estimand** is written only in terms of things we can observe:

$$\sum_{x} \Big(E[Y \mid D{=}1, X{=}x] - E[Y \mid D{=}0, X{=}x] \Big) \, P(X{=}x)$$

where D is the treatment, Y the outcome, and X a set of observed covariates

From theoretical to empirical estimand

The **empirical estimand** is written only in terms of things we can observe:

$$\sum_{x} \Big(E[Y \mid D{=}1, X{=}x] - E[Y \mid D{=}0, X{=}x] \Big) \, P(X{=}x)$$

where D is the treatment, Y the outcome, and X a set of observed covariates

- It equals the theoretical estimand **only under assumptions**: here, that X captures every reason why treated and untreated students differ.

From theoretical to empirical estimand

The **empirical estimand** is written only in terms of things we can observe:

$$\sum_{x} \Big(E[Y \mid D{=}1, X{=}x] - E[Y \mid D{=}0, X{=}x] \Big) \, P(X{=}x)$$

where D is the treatment, Y the outcome, and X a set of observed covariates

- It equals the theoretical estimand **only under assumptions**: here, that X captures every reason why treated and untreated students differ.
- Making that link is the **Link** step: its **identification assumptions** are what a DAG makes explicit (after the break).

From theoretical to empirical estimand

The **empirical estimand** is written only in terms of things we can observe:

$$\sum_{x} \Big(E[Y \mid D{=}1, X{=}x] - E[Y \mid D{=}0, X{=}x] \Big) \, P(X{=}x)$$

where D is the treatment, Y the outcome, and X a set of observed covariates

- It equals the theoretical estimand **only under assumptions**: here, that X captures every reason why treated and untreated students differ.
- Making that link is the **Link** step: its **identification assumptions** are what a DAG makes explicit (after the break).
- The **Learn** step, estimation (OLS, matching, weighting…), comes **last**.

Your turn ✏️ (~3 min, with the person next to you)

Pick **one** question and write down its **unit-specific quantity** and **target population**.
Is it descriptive or causal?

1. Does living with an out-group roommate reduce prejudice?
2. Do anonymous CVs reduce hiring discrimination?
3. Do protected bike lanes increase cycling?

Your turn ✏️ (~3 min, with the person next to you)

Pick **one** question and write down its **unit-specific quantity** and **target population**.
Is it descriptive or causal?

1. Does living with an out-group roommate reduce prejudice?
2. Do anonymous CVs reduce hiring discrimination?
3. Do protected bike lanes increase cycling?

Would a regression of Y on the treatment plus controls give you that quantity? Why (not)?

Break

10 minutes

DAGs: drawing your assumptions

What a DAG says, and what it doesn’t

What a DAG says, and what it doesn't

- **Nodes** are variables, observed or **unobserved**.

What a DAG says, and what it doesn't

- **Nodes** are variables, observed or **unobserved**.
- An **arrow** says "may cause". It makes no claim about size or sign (*i.e., non-parametric*).

What a DAG says, and what it doesn't

- **Nodes** are variables, observed or **unobserved**.
- An **arrow** says "may cause". It makes no claim about size or sign (*i.e., non-parametric*).
- A **missing arrow** is a strong claim: **no direct effect at all**.

What a DAG says, and what it doesn't

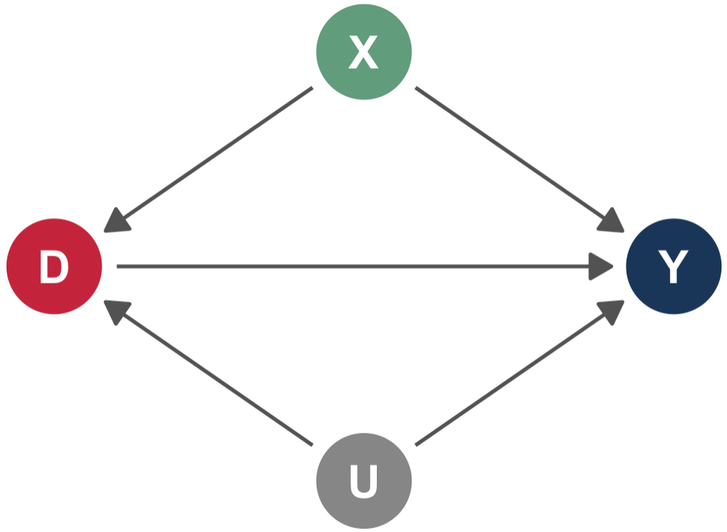
- **Nodes** are variables, observed or **unobserved**.
- An **arrow** says "may cause". It makes no claim about size or sign (*i.e., non-parametric*).
- A **missing arrow** is a strong claim: **no direct effect at all**.
- **Acyclic**: no variable causes itself through a loop. Time moves forward.

What a DAG says, and what it doesn't

- **Nodes** are variables, observed or **unobserved**.
- An **arrow** says "may cause". It makes no claim about size or sign (*i.e., non-parametric*).
- A **missing arrow** is a strong claim: **no direct effect at all**.
- **Acyclic**: no variable causes itself through a loop. Time moves forward.
- Drawn from **theory and knowledge**, not from the data.

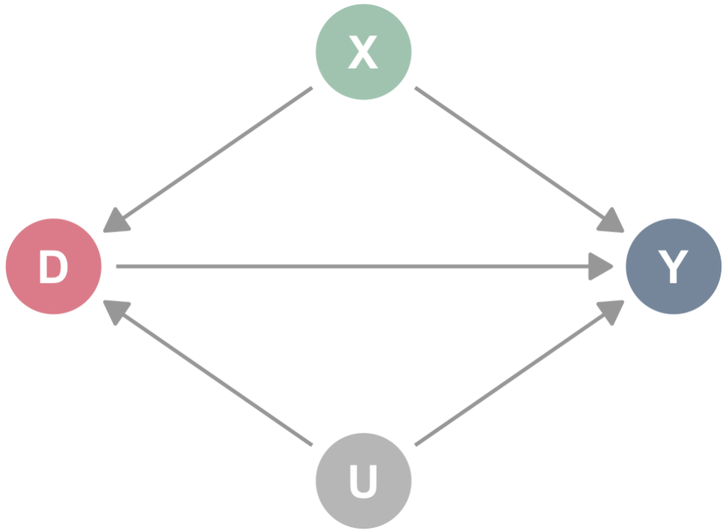
What a DAG says, and what it doesn't

- **Nodes** are variables, observed or **unobserved**.
- An **arrow** says "may cause". It makes no claim about size or sign (*i.e., non-parametric*).
- A **missing arrow** is a strong claim: **no direct effect at all**.
- **Acyclic**: no variable causes itself through a loop. Time moves forward.
- Drawn from **theory and knowledge**, not from the data.



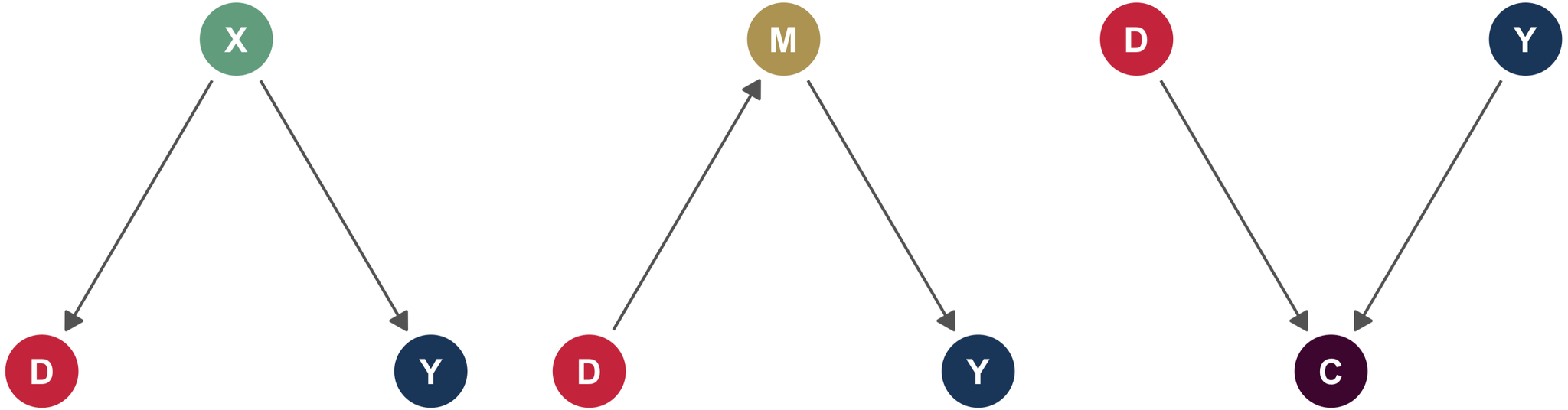
What a DAG says, and what it doesn't

- **Nodes** are variables, observed or **unobserved**.
- An **arrow** says "may cause". It makes no claim about size or sign (*i.e., non-parametric*).
- A **missing arrow** is a strong claim: **no direct effect at all**.
- **Acyclic**: no variable causes itself through a loop. Time moves forward.
- Drawn from **theory and knowledge**, not from the data.



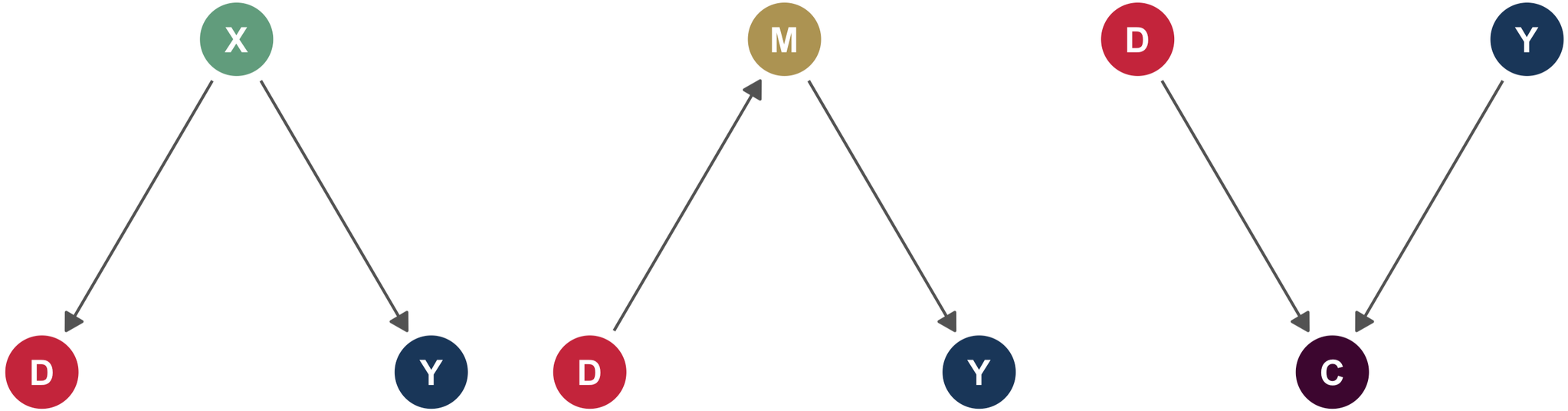
Two languages for the same problem, with different strengths: DAGs make complex assumptions **visible**; potential outcomes tie them to **assignment, heterogeneity, and estimation** (Imbens 2020).

Three building blocks: confounders, mediators, colliders



In the DAG literature, these three structures are also called a *fork*, a *chain*, and an *inverted fork* (or collider).

Three building blocks: confounders, mediators, colliders

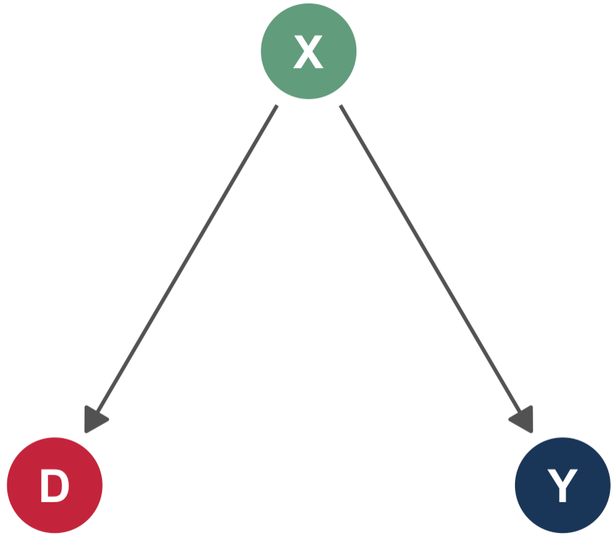


X is a **confounder**

D ← X → Y

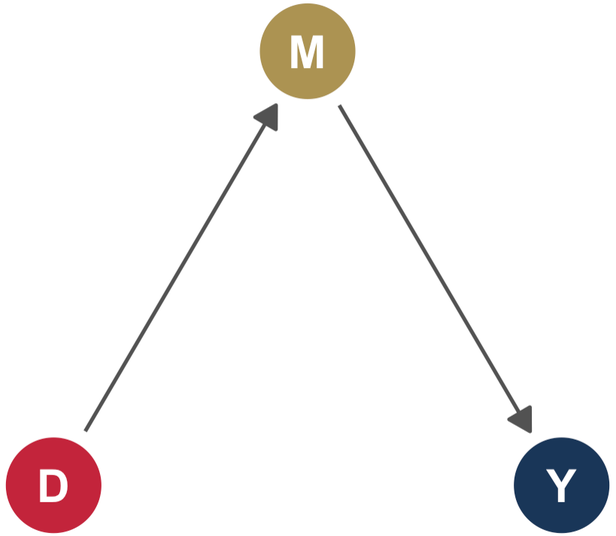
In the DAG literature, these three structures are also called a *fork*, a *chain*, and an *inverted fork* (or collider).

Three building blocks: confounders, mediators, colliders



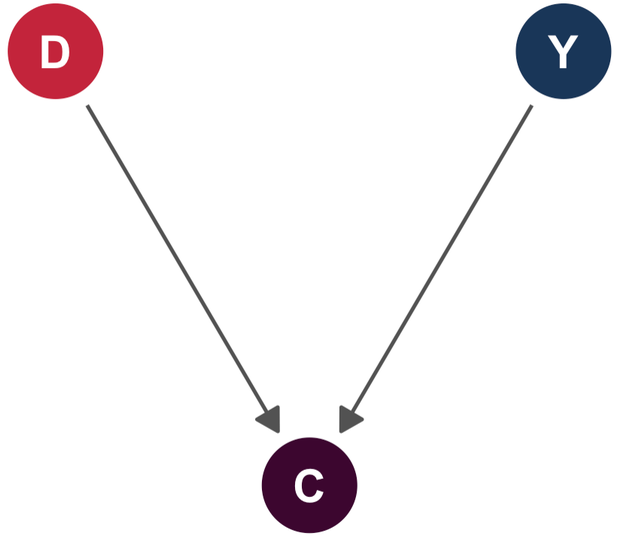
X is a **confounder**

D ← X → Y



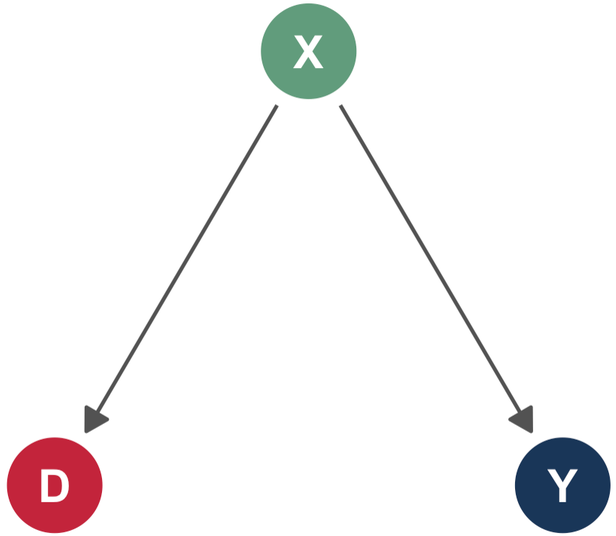
M is a **mediator**

D → M → Y



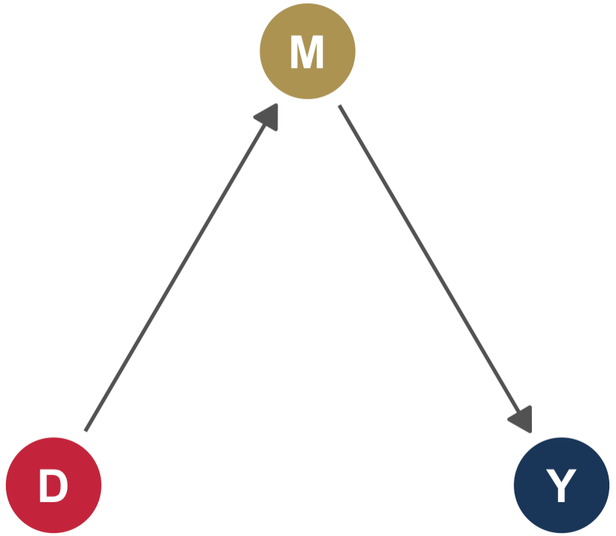
In the DAG literature, these three structures are also called a *fork*, a *chain*, and an *inverted fork* (or collider).

Three building blocks: confounders, mediators, colliders



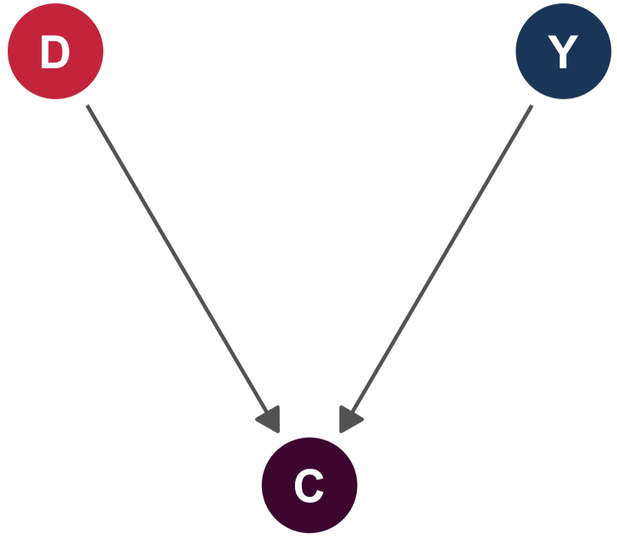
X is a **confounder**

D ← X → Y



M is a **mediator**

D → M → Y



C is a **collider**

D → C ← Y

In the DAG literature, these three structures are also called a *fork*, a *chain*, and an *inverted fork* (or collider).

Two rules for paths

Two rules for paths

A **path** is any sequence of arrows between two nodes, whatever their direction. An open path transmits association.

Two rules for paths

A **path** is any sequence of arrows between two nodes, whatever their direction. An open path transmits association.

Rule 1

Paths through a confounder or a mediator are **open**.
Conditioning on the confounder or mediator **closes** the path.

Two rules for paths

A **path** is any sequence of arrows between two nodes, whatever their direction. An open path transmits association.

Rule 1

Paths through a confounder or a mediator are **open**.
Conditioning on the confounder or mediator **closes** the path.

Rule 2

Colliders are **closed**.
Conditioning on a collider, or on anything it causes, **opens** the path.

Two rules for paths

A **path** is any sequence of arrows between two nodes, whatever their direction. An open path transmits association.

Rule 1

Paths through a confounder or a mediator are **open**.
Conditioning on the confounder or mediator **closes** the path.

Rule 2

Colliders are **closed**.
Conditioning on a collider, or on anything it causes, **opens** the path.

The association is **causal** when the path follows the arrows from D to Y: directly ($D \rightarrow Y$) or through a **mediator** ($D \rightarrow M \rightarrow Y$).

It is **non-causal** when the path runs back through a **confounder** ($D \leftarrow X \rightarrow Y$) or through a **collider** we conditioned on ($D \rightarrow C \leftarrow Y$).

Two rules for paths

A **path** is any sequence of arrows between two nodes, whatever their direction. An open path transmits association.

Rule 1

Paths through a confounder or a mediator are **open**.
Conditioning on the confounder or mediator **closes** the path.

Rule 2

Colliders are **closed**.
Conditioning on a collider, or on anything it causes, **opens** the path.

The association is **causal** when the path follows the arrows from D to Y: directly ($D \rightarrow Y$) or through a **mediator** ($D \rightarrow M \rightarrow Y$).

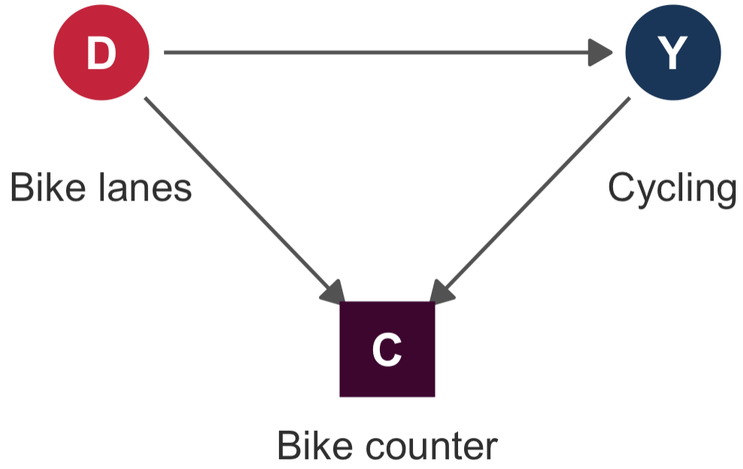
It is **non-causal** when the path runs back through a **confounder** ($D \leftarrow X \rightarrow Y$) or through a **collider** we conditioned on ($D \rightarrow C \leftarrow Y$).

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

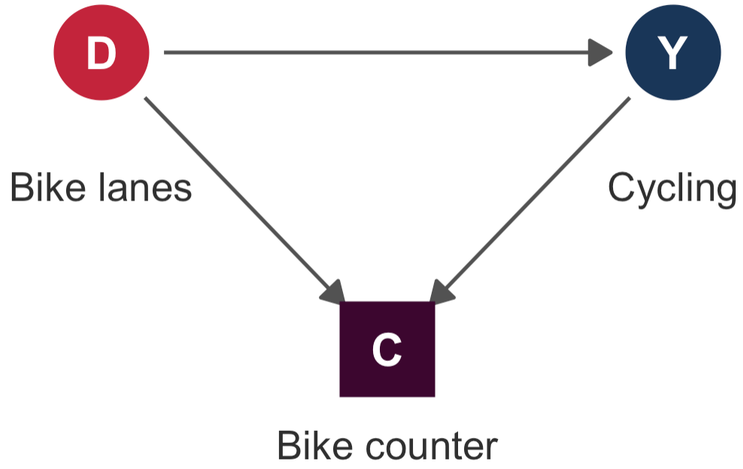


Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

- Do **protected bike lanes** increase **cycling**? We only have data from districts with automatic **bike counters**.

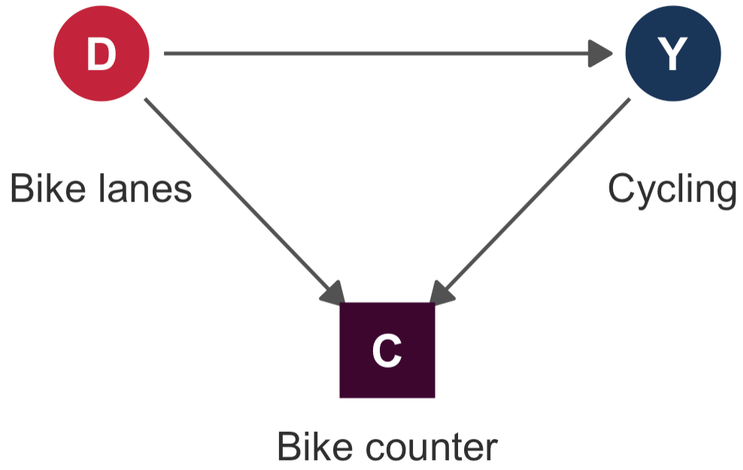


Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

- Do **protected bike lanes** increase **cycling**? We only have data from districts with automatic **bike counters**.
- We have reason to believe that cities install counters where lanes exist **or** where cycling is already high: the counter is a **collider**.

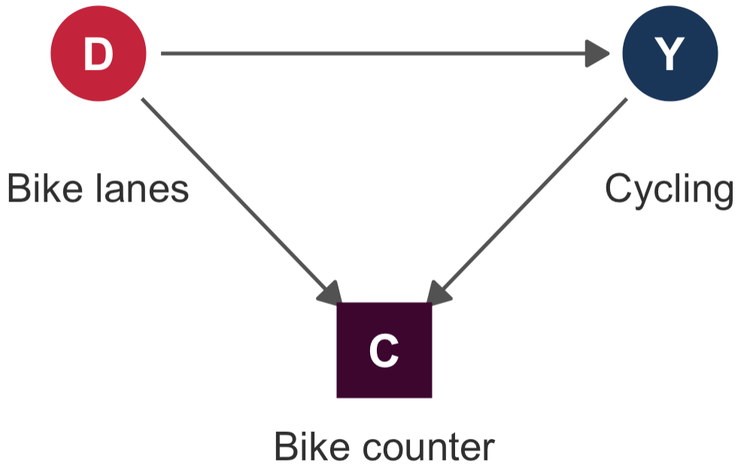


Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

- Do **protected bike lanes** increase **cycling**? We only have data from districts with automatic **bike counters**.
- We have reason to believe that cities install counters where lanes exist **or** where cycling is already high: the counter is a **collider**.
- In the sample, a district **without** lanes usually got its counter because cycling was already high.

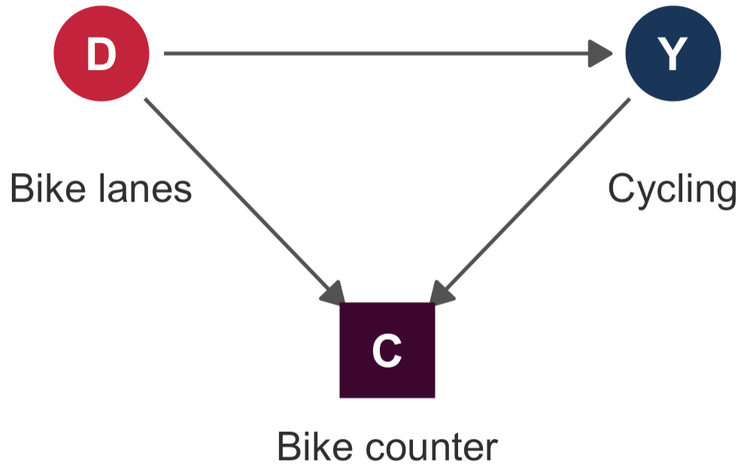


Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

- Do **protected bike lanes** increase **cycling**? We only have data from districts with automatic **bike counters**.
- We have reason to believe that cities install counters where lanes exist **or** where cycling is already high: the counter is a **collider**.
- In the sample, a district **without** lanes usually got its counter because cycling was already high.
- So no-lane districts look like heavy-cycling places: lanes and cycling become **negatively associated** among counter districts.

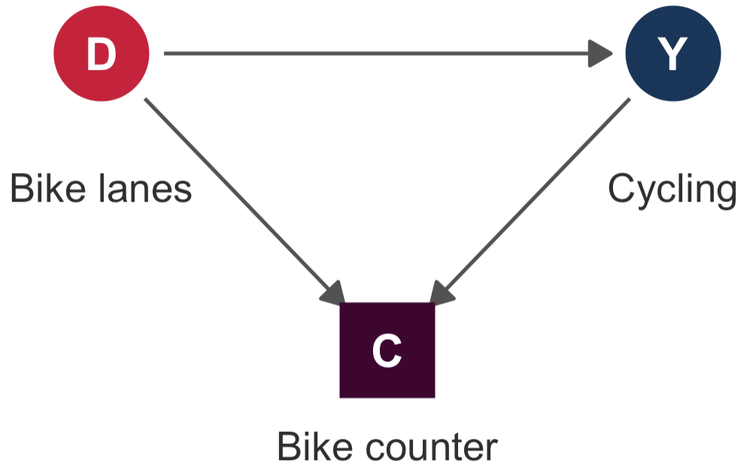


Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

- Do **protected bike lanes** increase **cycling**? We only have data from districts with automatic **bike counters**.
- We have reason to believe that cities install counters where lanes exist **or** where cycling is already high: the counter is a **collider**.
- In the sample, a district **without** lanes usually got its counter because cycling was already high.
- So no-lane districts look like heavy-cycling places: lanes and cycling become **negatively associated** among counter districts.
- The lane estimate is biased **downward**. It can shrink, vanish, or flip sign, even if lanes truly raise cycling.

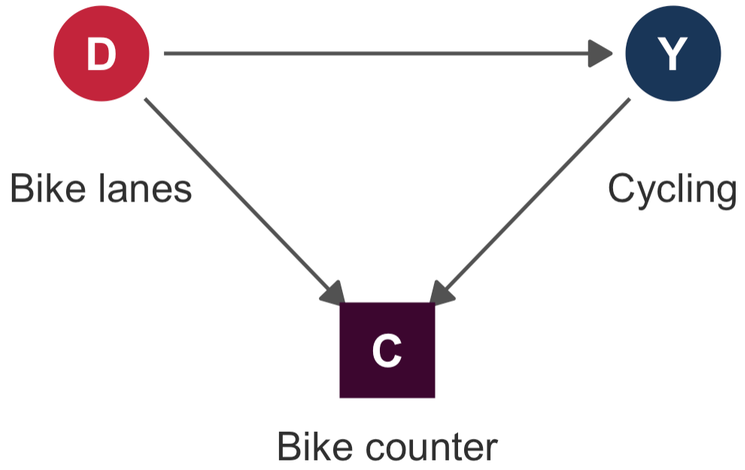


Conditioning on a collider: the bike-counter problem

"Conditioning" = controlling in a regression, stratifying, matching, **or selecting the sample**.

Fair warning: I am very much into cycling (and a bit unidimensional), so expect bike stuff to keep popping up in this course.

- Do **protected bike lanes** increase **cycling**? We only have data from districts with automatic **bike counters**.
- We have reason to believe that cities install counters where lanes exist **or** where cycling is already high: the counter is a **collider**.
- In the sample, a district **without** lanes usually got its counter because cycling was already high.
- So no-lane districts look like heavy-cycling places: lanes and cycling become **negatively associated** among counter districts.
- The lane estimate is biased **downward**. It can shrink, vanish, or flip sign, even if lanes truly raise cycling.
- More counter districts do not help: the bias comes from **who is in the data**, not from sample size.



The backdoor criterion

The backdoor criterion

A set of variables Z identifies the effect of D on Y if

1. no variable in Z is a **descendant** of D, and
2. Z **blocks every backdoor path**: every path between D and Y that starts with an arrow **into** D.

The backdoor criterion

> A set of variables Z identifies the effect of D on Y if
>
> 1. no variable in Z is a **descendant** of D, and
> 2. Z **blocks every backdoor path**: every path between D and Y that starts with an arrow **into** D.

Recipe

The backdoor criterion

> A set of variables Z identifies the effect of D on Y if
>
> 1. no variable in Z is a **descendant** of D, and
> 2. Z **blocks every backdoor path**: every path between D and Y that starts with an arrow **into** D.

Recipe

1. Draw the DAG for your estimand.

The backdoor criterion

> A set of variables Z identifies the effect of D on Y if
>
> 1. no variable in Z is a **descendant** of D, and
> 2. Z **blocks every backdoor path**: every path between D and Y that starts with an arrow **into** D.

Recipe

1. Draw the DAG for your estimand.
2. List all paths from D to Y.

The backdoor criterion

> A set of variables Z identifies the effect of D on Y if
>
> 1. no variable in Z is a **descendant** of D, and
> 2. Z **blocks every backdoor path**: every path between D and Y that starts with an arrow **into** D.

Recipe

1. Draw the DAG for your estimand.
2. List all paths from D to Y.
3. Mark the backdoor paths, and whether each is open or closed.

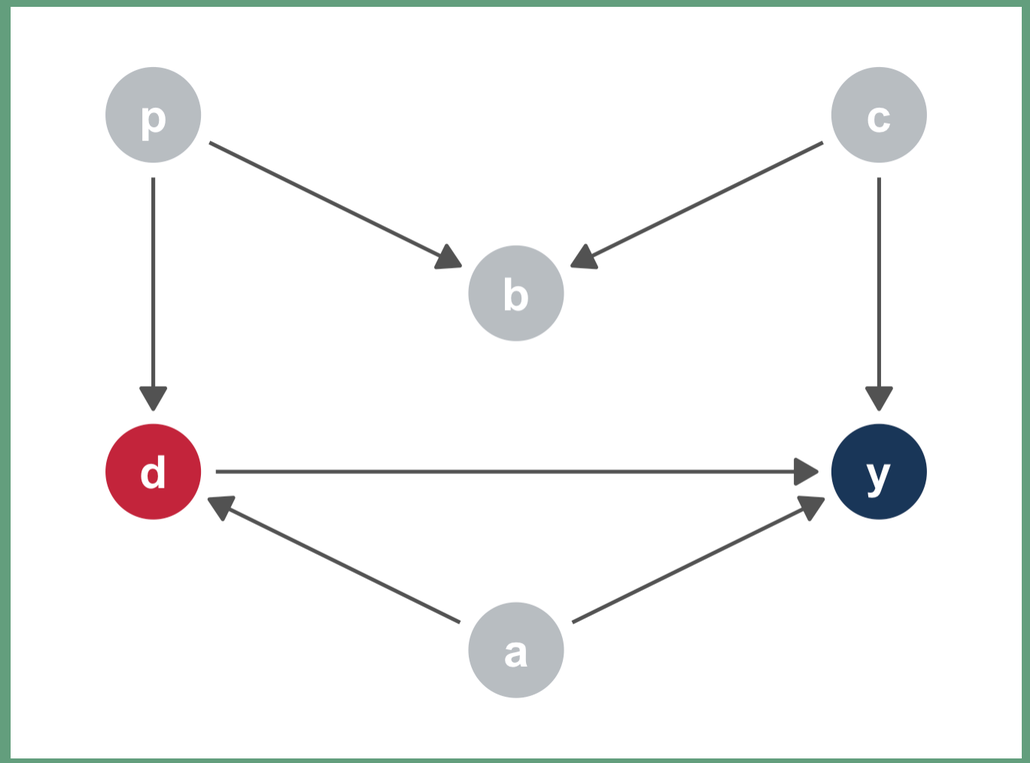
The backdoor criterion

> A set of variables Z identifies the effect of D on Y if
>
> 1. no variable in Z is a **descendant** of D, and
> 2. Z **blocks every backdoor path**: every path between D and Y that starts with an arrow **into** D.

Recipe

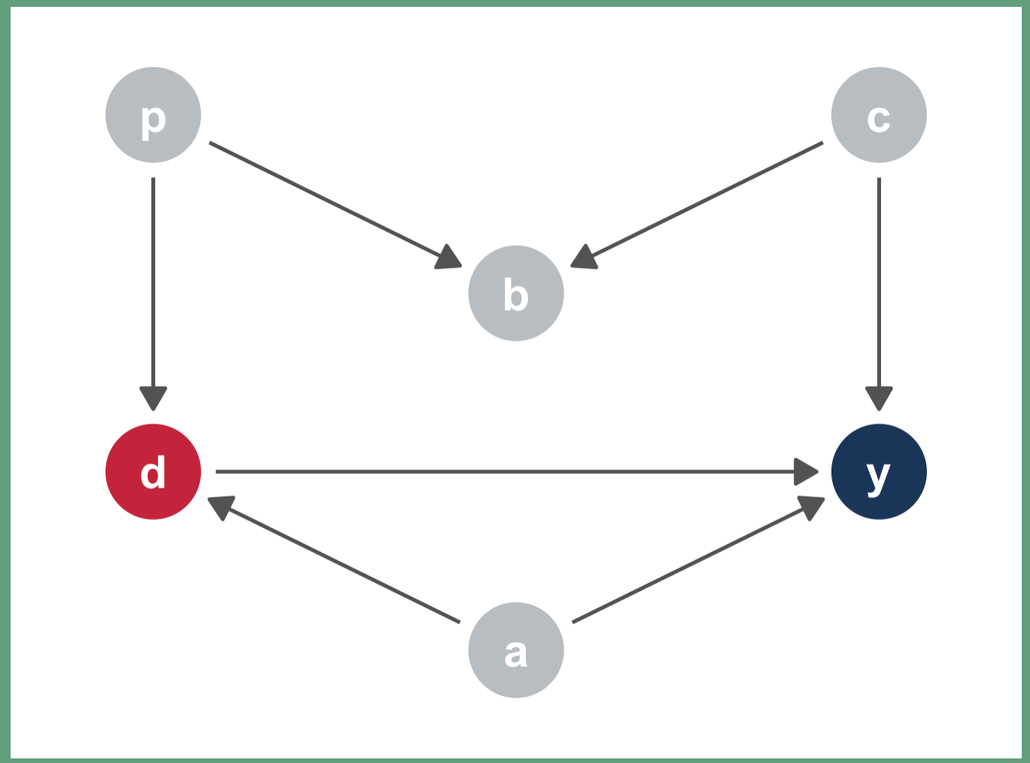
1. Draw the DAG for your estimand.
2. List all paths from D to Y.
3. Mark the backdoor paths, and whether each is open or closed.
4. Find a set that closes the open ones **without** opening new ones (Rule 2!).

Let's do this together 🔍



1. Identify the **causal** and **non-causal** paths from d to y.
2. Are the non-causal paths **naturally closed** (by a collider)?
3. What should we **condition on** to close the open non-causal paths?

Let's do this together 🔍



1. Identify the **causal** and **non-causal** paths from d to y.
2. Are the non-causal paths **naturally closed** (by a collider)?
3. What should we **condition on** to close the open non-causal paths?

A colleague controls for b "to be safe". What else must they add?

Good and bad controls

More controls is not safer

Cinelli, Forney & Pearl (2024), "A crash course in good and bad controls", *Sociological Methods & Research* 53(3)

More controls is not safer

Good: removes bias

Confounders, and proxies of unobserved confounders, that sit on open backdoor paths

Cinelli, Forney & Pearl (2024), "A crash course in good and bad controls", *Sociological Methods & Research* 53(3)

More controls is not safer

Good: removes bias

Confounders, and proxies of unobserved confounders, that sit on open backdoor paths

Neutral, but helps precision

Causes of the outcome only (e.g., baseline prejudice): smaller standard errors, no bias

Cinelli, Forney & Pearl (2024), "A crash course in good and bad controls", *Sociological Methods & Research* 53(3)

More controls is not safer

Good: removes bias

Confounders, and proxies of unobserved confounders, that sit on open backdoor paths

Neutral, but helps precision

Causes of the outcome only (e.g., baseline prejudice): smaller standard errors, no bias

Neutral, but hurts precision

Causes of the treatment only (e.g., dorm-allocation lottery number): larger standard errors, and they amplify any remaining bias

Cinelli, Forney & Pearl (2024), "A crash course in good and bad controls", *Sociological Methods & Research* 53(3)

More controls is not safer

Good: removes bias

Confounders, and proxies of unobserved confounders, that sit on open backdoor paths

Neutral, but helps precision

Causes of the outcome only (e.g., baseline prejudice): smaller standard errors, no bias

Neutral, but hurts precision

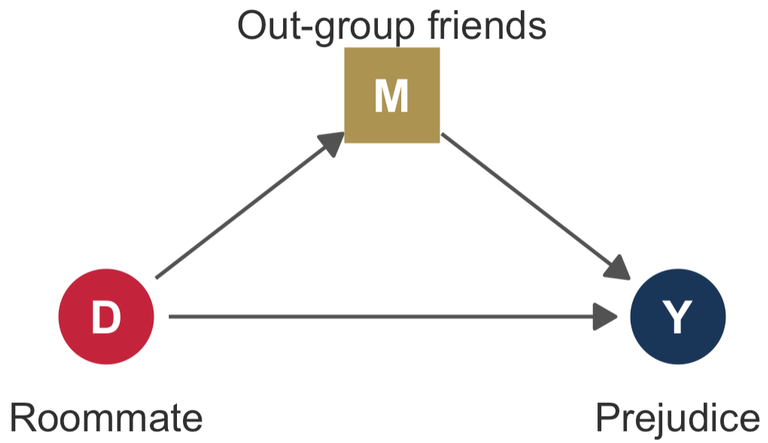
Causes of the treatment only (e.g., dorm-allocation lottery number): larger standard errors, and they amplify any remaining bias

Bad: creates bias

Mediators, colliders and their descendants, descendants of the treatment, and "M-bias" pre-treatment colliders

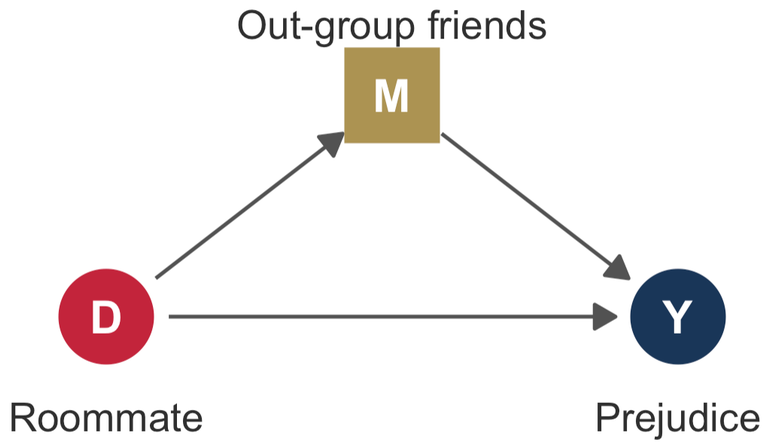
Cinelli, Forney & Pearl (2024), "A crash course in good and bad controls", *Sociological Methods & Research* 53(3)

Bad control 1: controlling away the effect



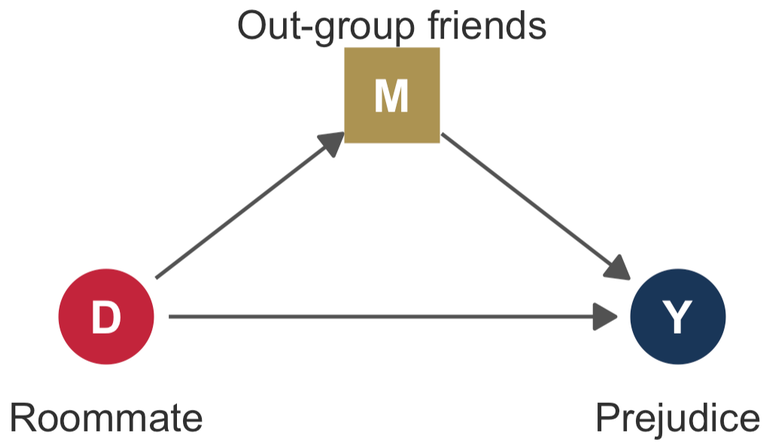
Bad control 1: controlling away the effect

- Contact may reduce prejudice **through** friendships with out-group members.



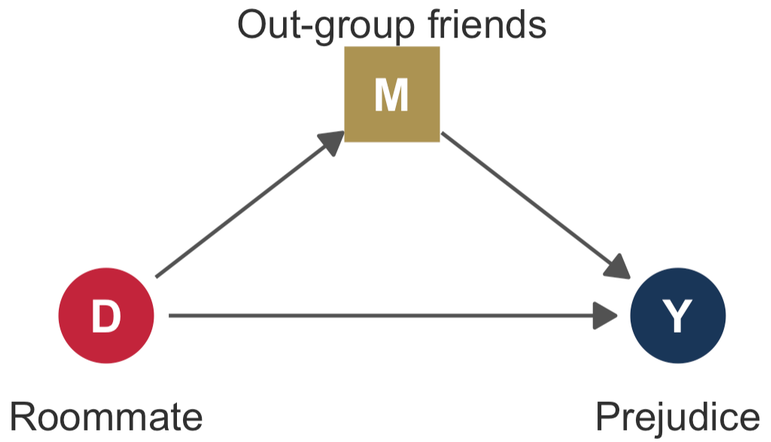
Bad control 1: controlling away the effect

- Contact may reduce prejudice **through** friendships with out-group members.
- Controlling for **out-group friendships** at follow-up blocks that path.



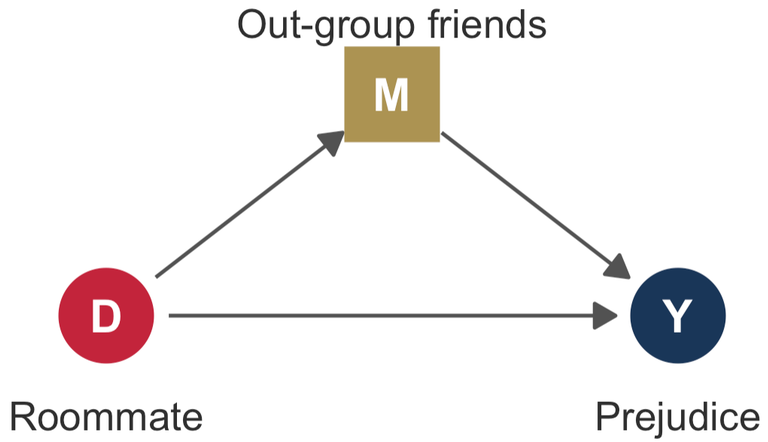
Bad control 1: controlling away the effect

- Contact may reduce prejudice **through** friendships with out-group members.
- Controlling for **out-group friendships** at follow-up blocks that path.
- The coefficient on roommate type then shows only what is left: **not the total effect**.

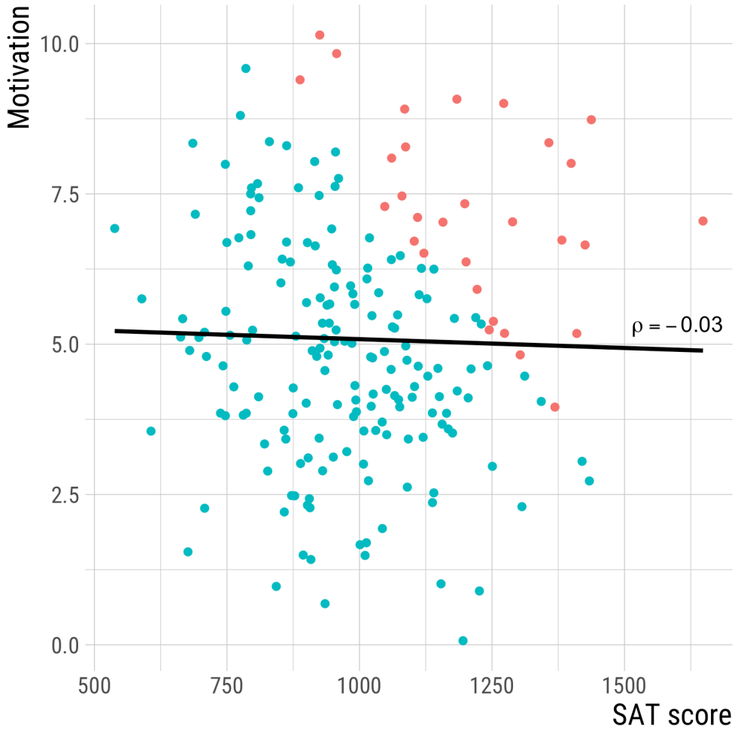


Bad control 1: controlling away the effect

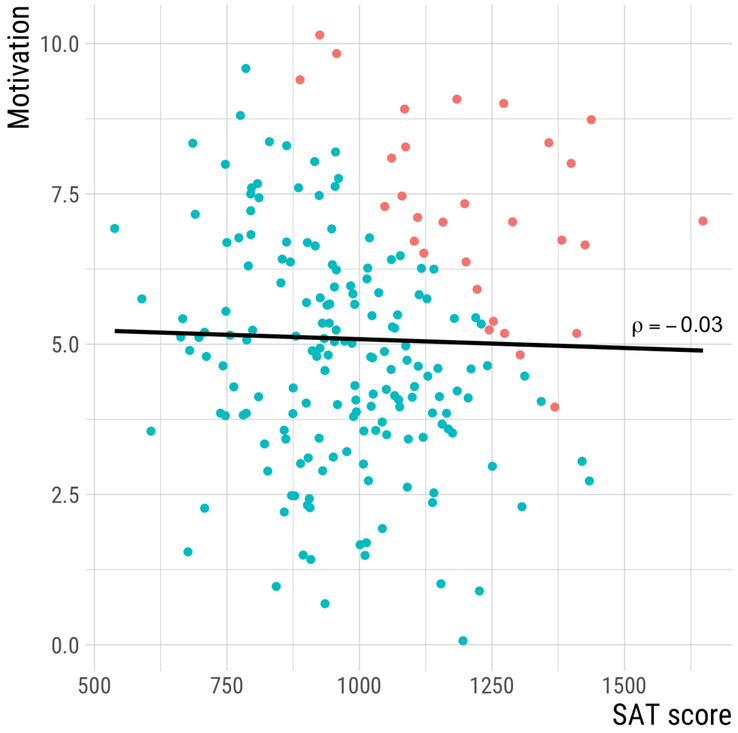
- Contact may reduce prejudice **through** friendships with out-group members.
- Controlling for **out-group friendships** at follow-up blocks that path.
- The coefficient on roommate type then shows only what is left: **not the total effect**.
- Same problem with any **post-treatment** variable, including in experiments: controlling for it, dropping people on it, or subsetting on it (Montgomery, Nyhan & Torres 2018).

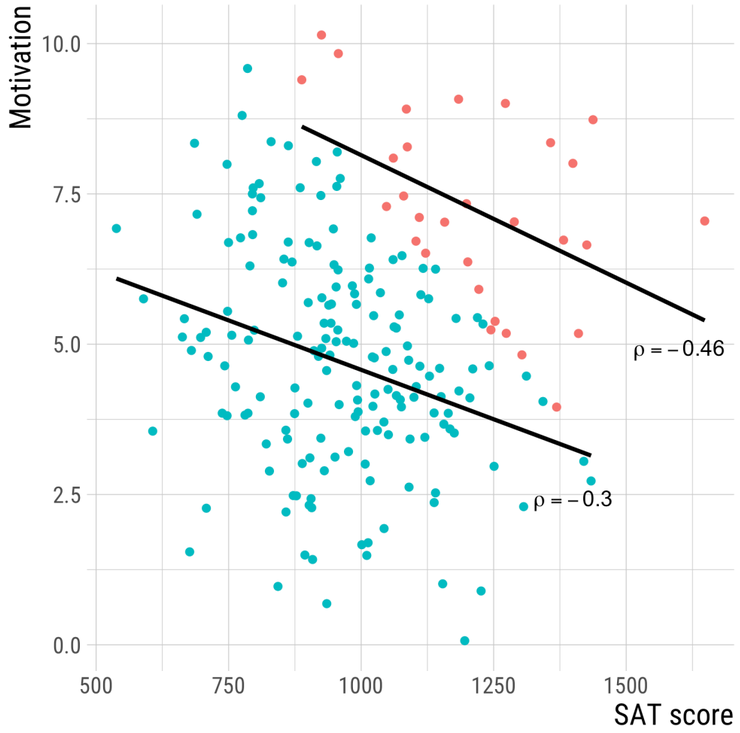


Bad control 2: colliders manufacture correlation

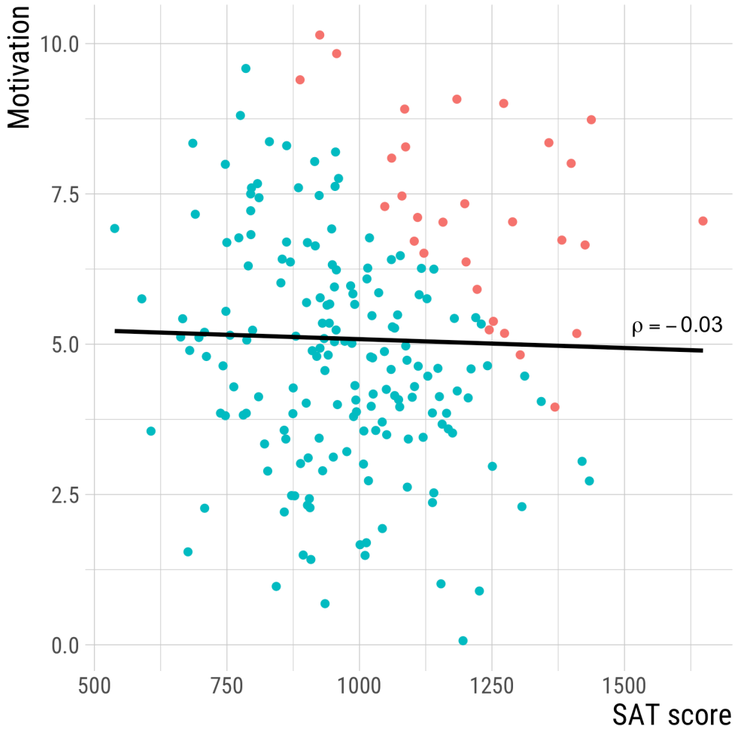


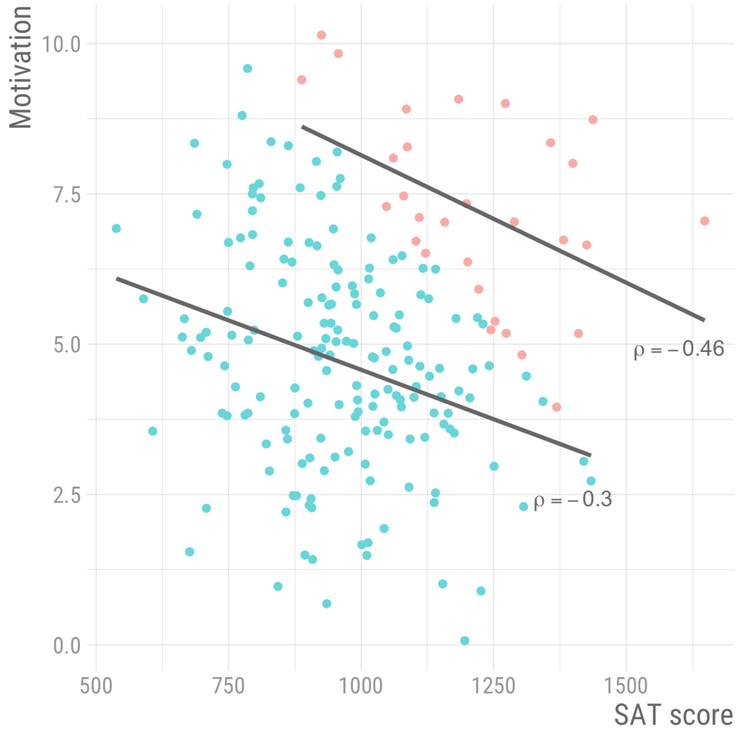
Bad control 2: colliders manufacture correlation





Bad control 2: colliders manufacture correlation





SAT score and motivation are independent, but both get you **admitted**. **Among admitted students**, they look negatively related.

In the lab on Monday, you will simulate both mistakes:

Skip a confounder: shoe size “predicts” salary, until you control for **sex**.

Control for a collider: beauty and talent look negatively related, once you condition on **celebrity**.

In the lab on Monday, you will simulate both mistakes:

Skip a confounder: shoe size "predicts" salary, until you control for **sex**.

Control for a collider: beauty and talent look negatively related, once you condition on **celebrity**.

These examples are silly on purpose.

In the lab on Monday, you will simulate both mistakes:

Skip a confounder: shoe size "predicts" salary, until you control for **sex**.

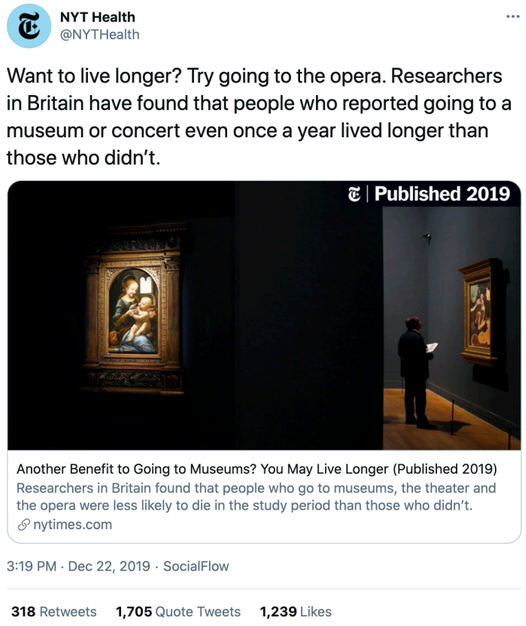
Control for a collider: beauty and talent look negatively related, once you condition on **celebrity**.

These examples are silly on purpose.

The things we study are not. Contact and prejudice, bike lanes and cycling, race and police force.

Ask yourself how obvious the **right model** would be there.

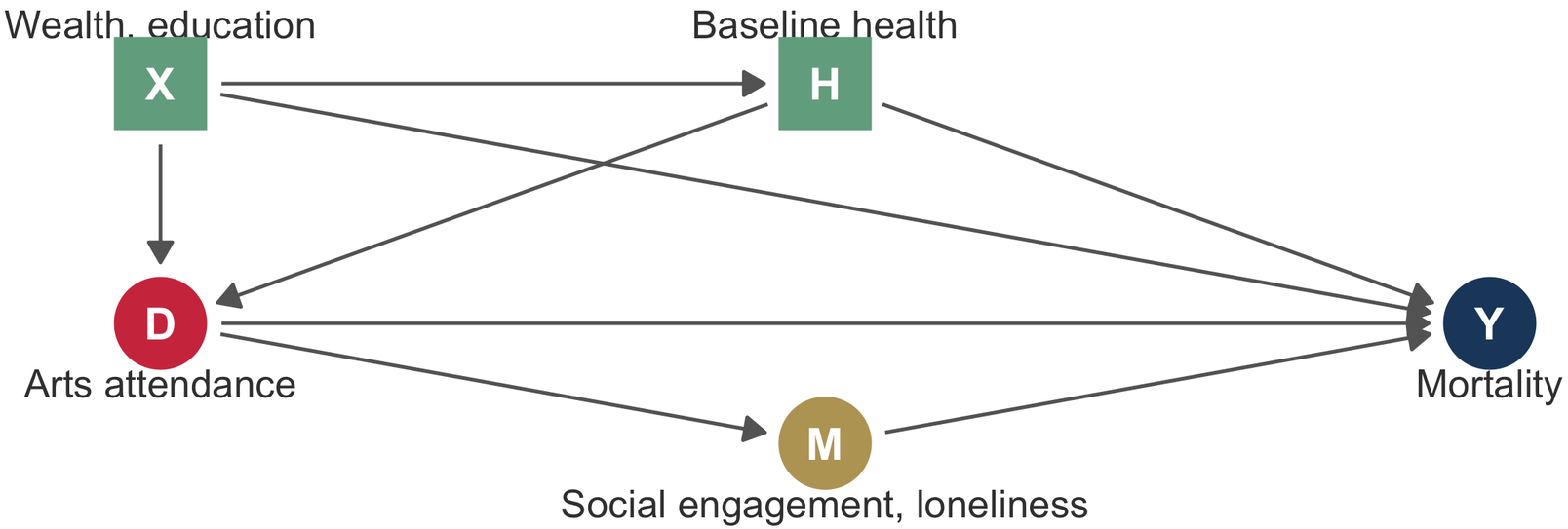
Opera, revisited: what did the authors control for?



> *"We adjusted models for **demographic variables** (age, sex, marital status, ethnicity, educational qualifications, wealth, employment status, and occupational status); **health related variables** (eyesight, hearing, depressive symptoms, other psychiatric conditions, diagnosis of cancer, lung disease or cardiovascular disease, history of any other long-term condition, smoking, alcohol consumption, sedentary behaviours, mobility, problems in undertaking activities of daily living, osteoporosis, and cognition); and **social covariates** (loneliness, number of close friends, living alone, frequency of civic engagement, frequency of social engagement, and whether participants had a hobby or pastime)."*

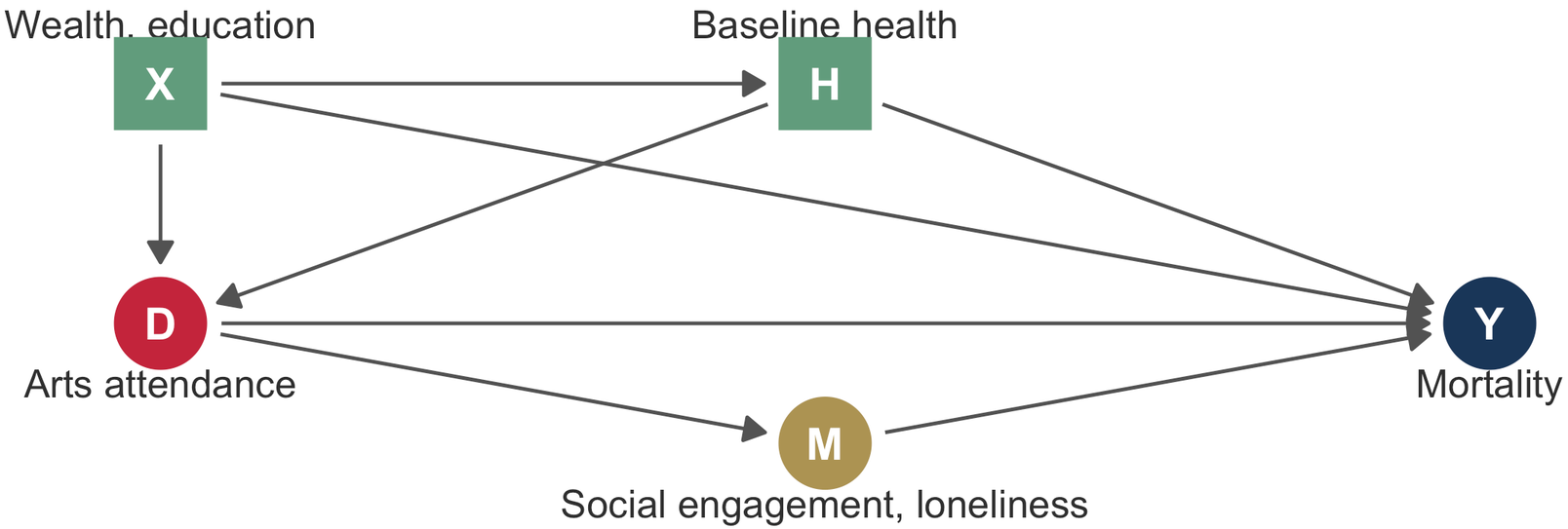
Opera, revisited: suppose this is the DAG

Not *the* true DAG, but **a DAG**: our theory of how arts attendance, health, wealth, and social life fit together, written down.



Opera, revisited: suppose this is the DAG

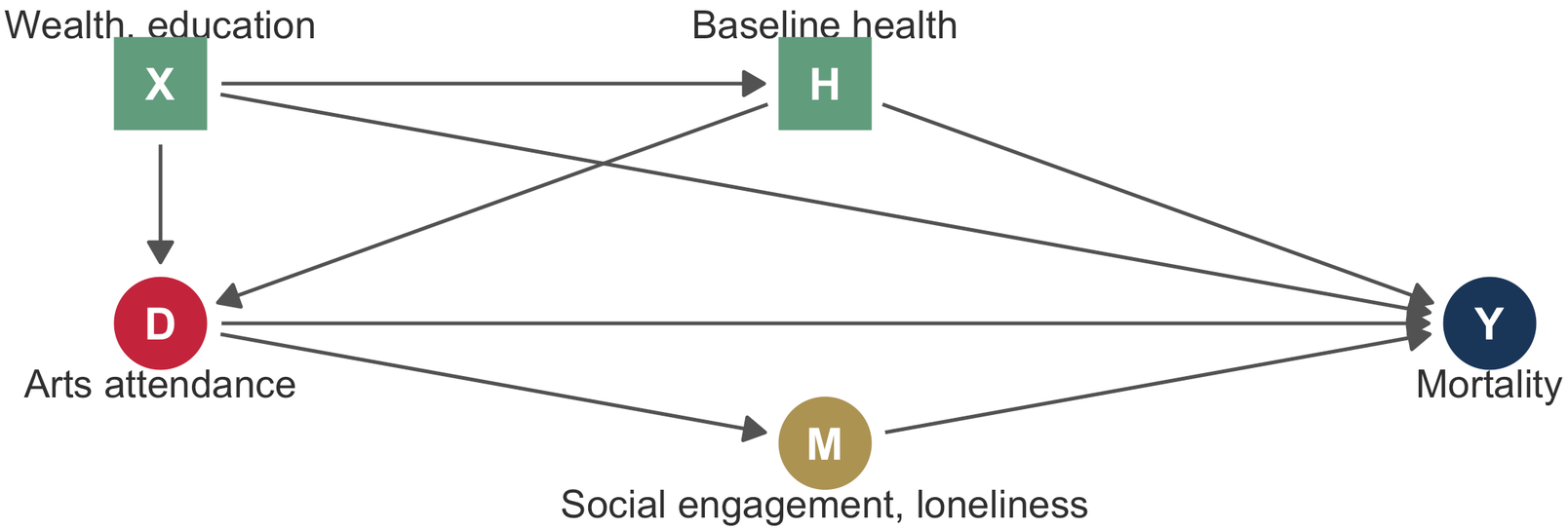
Not *the* true DAG, but **a DAG**: our theory of how arts attendance, health, wealth, and social life fit together, written down.



If this is the world: adjust for **wealth, education** and **baseline** health; leave **social engagement** out.

Opera, revisited: suppose this is the DAG

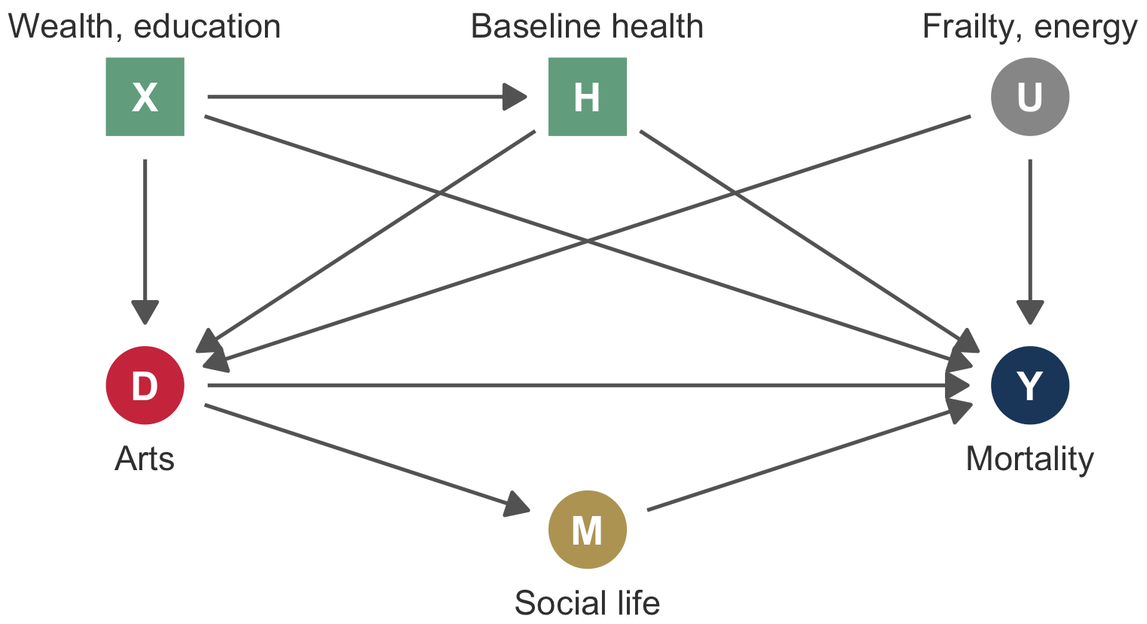
Not *the* true DAG, but **a DAG**: our theory of how arts attendance, health, wealth, and social life fit together, written down.



If this is the world: adjust for **wealth, education** and **baseline** health; leave **social engagement** out.

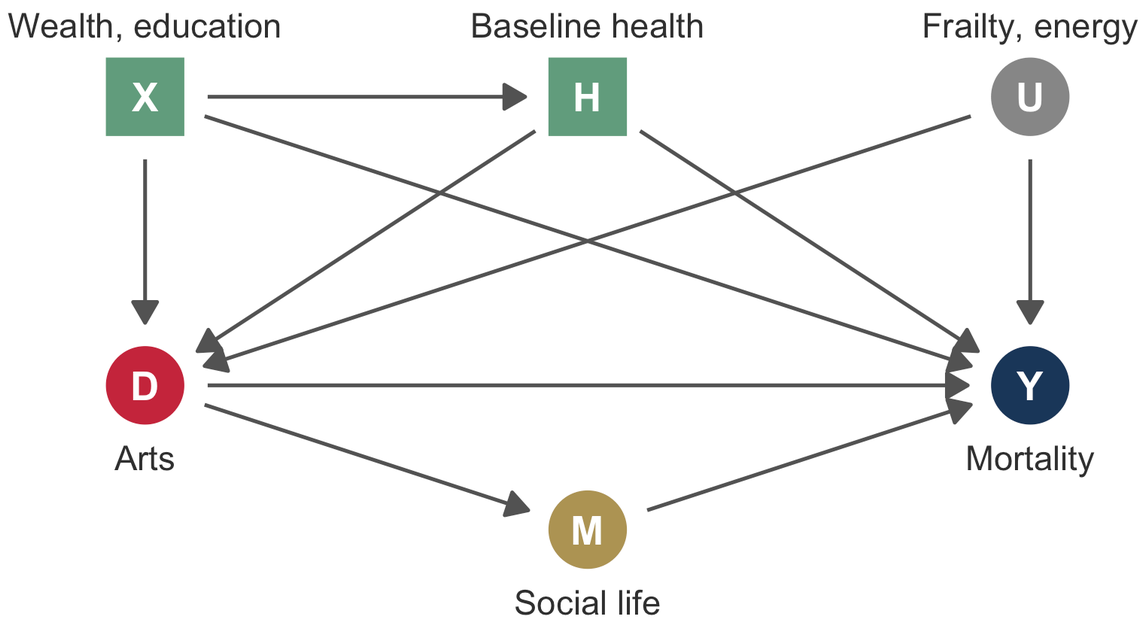
The leverage of a DAG: every arrow, and every **missing** arrow, is a claim **others can see and contest**.

Contest it: a critic’s DAG



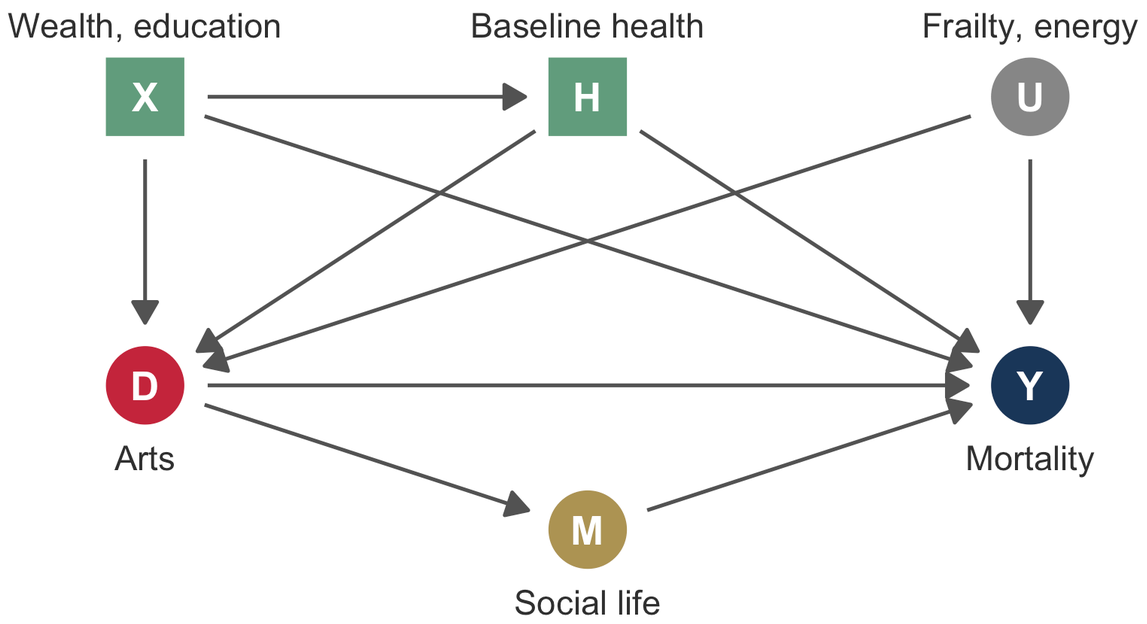
Contest it: a critic's DAG

- A critic: people with more **energy** and fewer **undiagnosed illnesses** both go out to the opera **and** live longer.



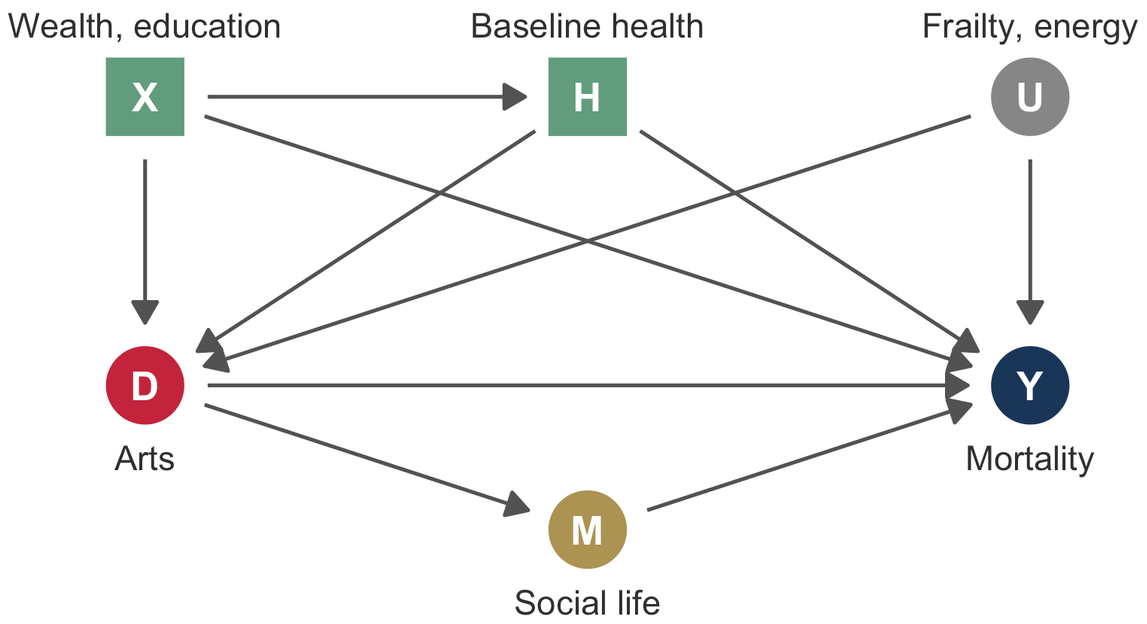
Contest it: a critic's DAG

- A critic: people with more **energy** and fewer **undiagnosed illnesses** both go out to the opera **and** live longer.
- Baseline health only partly measures this. The rest is an **unobserved** common cause, U.



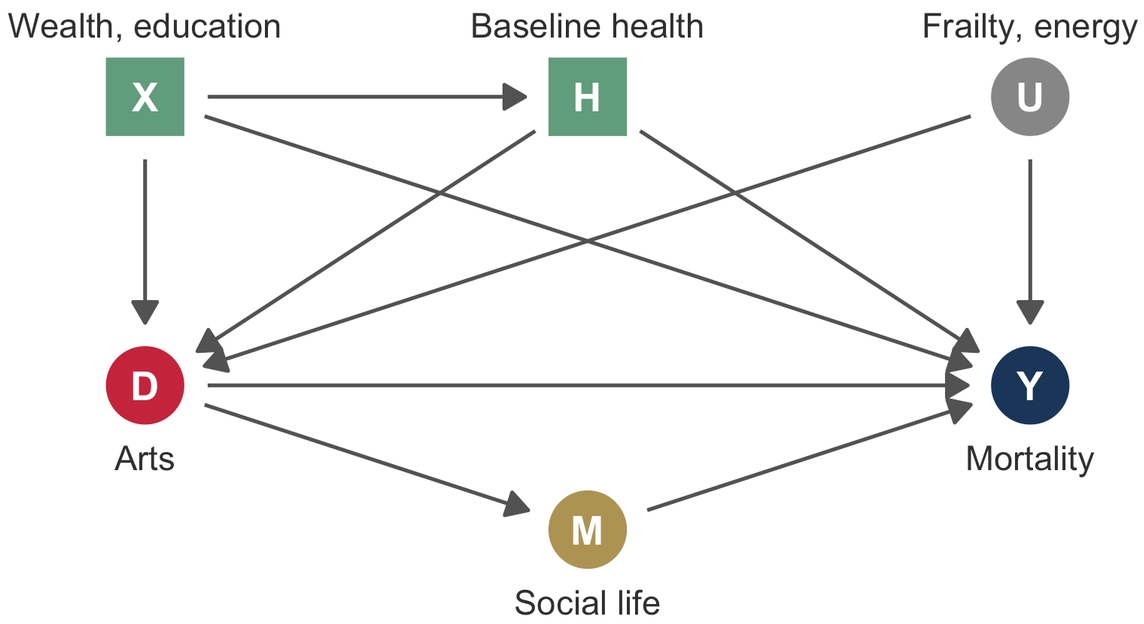
Contest it: a critic's DAG

- A critic: people with more **energy** and fewer **undiagnosed illnesses** both go out to the opera **and** live longer.
- Baseline health only partly measures this. The rest is an **unobserved** common cause, U.
- That adds a backdoor path, Arts ← U → Mortality, that **no measured control closes**.



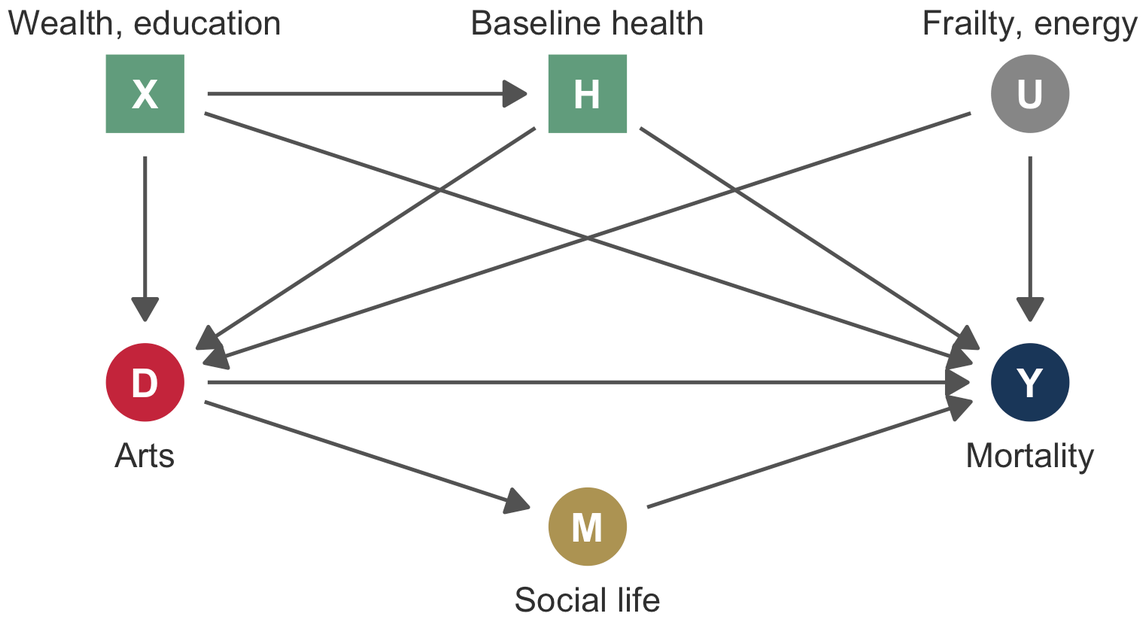
Contest it: a critic's DAG

- A critic: people with more **energy** and fewer **undiagnosed illnesses** both go out to the opera **and** live longer.
- Baseline health only partly measures this. The rest is an **unobserved** common cause, U.
- That adds a backdoor path, Arts $\leftarrow U \rightarrow$ Mortality, that **no measured control closes**.
- The disagreement is now **precise**: one arrow, which better health data or a different design could settle.



Contest it: a critic's DAG

- A critic: people with more **energy** and fewer **undiagnosed illnesses** both go out to the opera **and** live longer.
- Baseline health only partly measures this. The rest is an **unobserved** common cause, U.
- That adds a backdoor path, Arts $\leftarrow U \rightarrow$ Mortality, that **no measured control closes**.
- The disagreement is now **precise**: one arrow, which better health data or a different design could settle.



Disagreeing about a DAG is a **scientific debate, not a matter of taste**: you have to say which arrow, and why.

Where do your data come from?

Is there racial bias in police use of force?

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

Is there racial bias in police use of force?

- A question with enormous stakes, and one where **the data are produced by the very behavior under study**. The usual source: **administrative records** of police–civilian encounters.

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

Is there racial bias in police use of force?

- A question with enormous stakes, and one where **the data are produced by the very behavior under study**. The usual source: **administrative records** of police–civilian encounters.
- Previous research: Fryer (2019) analyzes police records with rich controls for context and civilian behavior.

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

Is there racial bias in police use of force?

- A question with enormous stakes, and one where **the data are produced by the very behavior under study**. The usual source: **administrative records** of police–civilian encounters.
- Previous research: Fryer (2019) analyzes police records with rich controls for context and civilian behavior.

Non-lethal force · NYC stop-and-frisk records

>50% more likely

for Black and Hispanic civilians; controls reduce the gap but do not eliminate it

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

Is there racial bias in police use of force?

- A question with enormous stakes, and one where **the data are produced by the very behavior under study**. The usual source: **administrative records** of police–civilian encounters.
- Previous research: Fryer (2019) analyzes police records with rich controls for context and civilian behavior.

Non-lethal force · NYC stop-and-frisk records

>50% more likely

for Black and Hispanic civilians; controls reduce the gap but do not eliminate it

Officer-involved shootings

no racial difference

in the raw data or with controls for context

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

Is there racial bias in police use of force?

- A question with enormous stakes, and one where **the data are produced by the very behavior under study**. The usual source: **administrative records** of police–civilian encounters.
- Previous research: Fryer (2019) analyzes police records with rich controls for context and civilian behavior.

Non-lethal force · NYC stop-and-frisk records

>50% more likely

for Black and Hispanic civilians; controls reduce the gap but do not eliminate it

Officer-involved shootings

no racial difference

in the raw data or with controls for context

The second finding became the headline: no bias in lethal force… but both comparisons are made **only among people already in a 'recorded' encounter**. If race shapes who ends up in those encounters, "no difference" may be exactly what discrimination looks like in the records.

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

Is there racial bias in police use of force?

- A question with enormous stakes, and one where **the data are produced by the very behavior under study**. The usual source: **administrative records** of police–civilian encounters.
- Previous research: Fryer (2019) analyzes police records with rich controls for context and civilian behavior.

Non-lethal force · NYC stop-and-frisk records

>50% more likely

for Black and Hispanic civilians; controls reduce the gap but do not eliminate it

Officer-involved shootings

no racial difference

in the raw data or with controls for context

The second finding became the headline: no bias in lethal force… but both comparisons are made **only among people already in a 'recorded' encounter**. If race shapes who ends up in those encounters, "no difference" may be exactly what discrimination looks like in the records.

- So before judging either number, ask two questions: **what is the estimand, and who is in the data?**

Fryer (2019), "An empirical analysis of racial differences in police use of force", *Journal of Political Economy* 127(3): 1210–1261

How does a row get into the data?

How does a row get into the data?

1. Encounter

An officer observes a civilian

not recorded

How does a row get into the data?

1. Encounter

An officer observes a civilian

not recorded

How does a row get into the data?

1. Encounter

An officer observes a civilian

not recorded

→

2. Stop or not?

The officer decides whom to stop

race may matter here

How does a row get into the data?

1. Encounter

An officer observes a civilian

not recorded

→

2. Stop or not?

The officer decides whom to stop

race may matter here

→

How does a row get into the data?

1. Encounter

An officer observes a civilian

not recorded

→

2. Stop or not?

The officer decides whom to stop

race may matter here

→

3. Force or not?

The officer decides how to act

and here

How does a row get into the data?

1. Encounter

An officer observes a civilian

not recorded

→

2. Stop or not?

The officer decides whom to stop

race may matter here

→

3. Force or not?

The officer decides how to act

and here

→

How does a row get into the data?

1. Encounter

An officer observes a civilian

not recorded

→

2. Stop or not?

The officer decides whom to stop

race may matter here

→

3. Force or not?

The officer decides how to act

and here

→

4. The record

One row per stop: the researcher's dataset

How does a row get into the data?

1. Encounter
An officer observes a civilian
not recorded

→

2. Stop or not?
The officer decides whom to stop
race may matter here

→

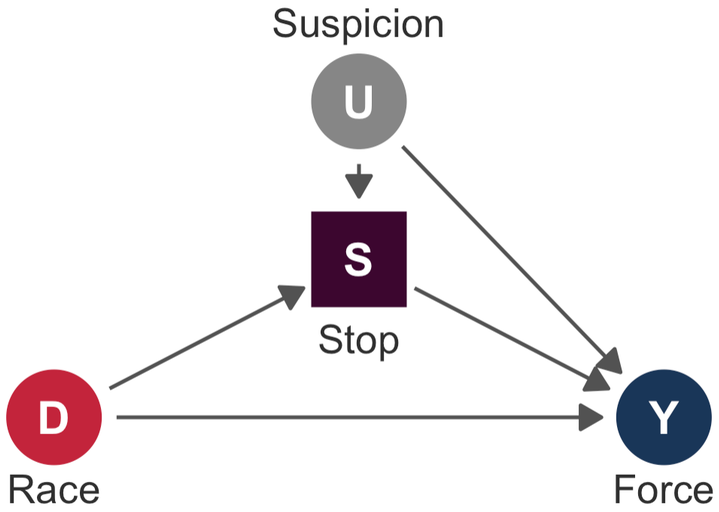
3. Force or not?
The officer decides how to act
and here

→

4. The record
One row per stop: the researcher's dataset

If race affects step 2, the people in the records are **already a selected group**, before anyone runs a regression.

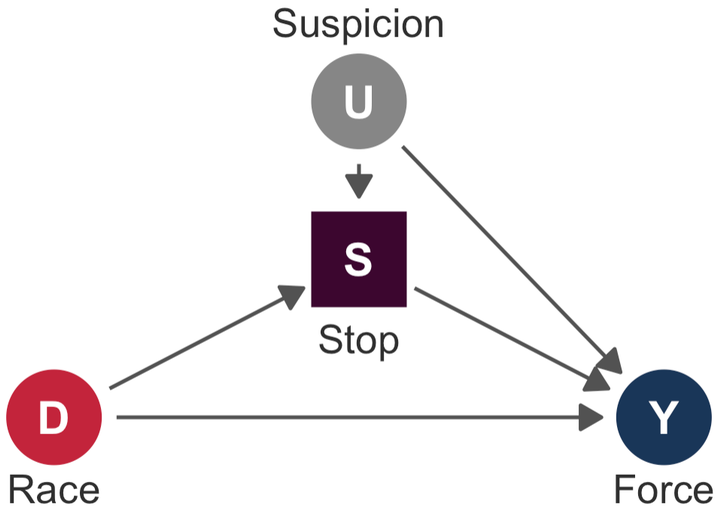
Draw the data-generating process as a DAG



Knox, Lowe & Mummolo (2020), *APSR* 114(3)

Draw the data-generating process as a DAG

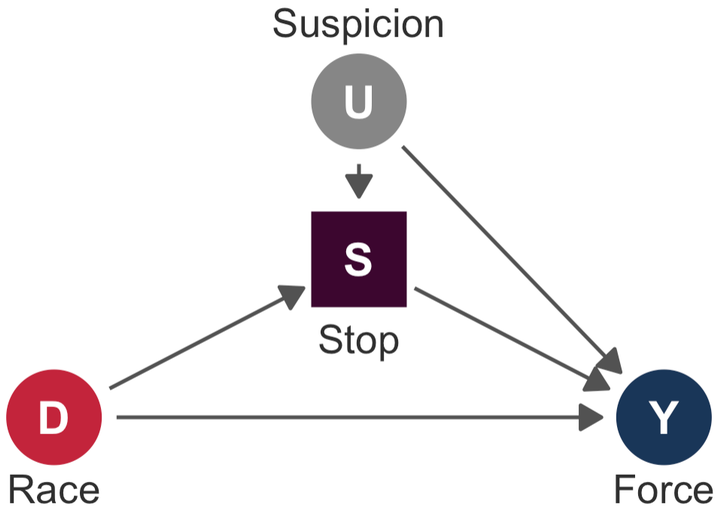
- **Race** (D) may affect whether someone is **stopped** (S), and whether **force** is used (Y)



Knox, Lowe & Mummolo (2020), *APSR* 114(3)

Draw the data-generating process as a DAG

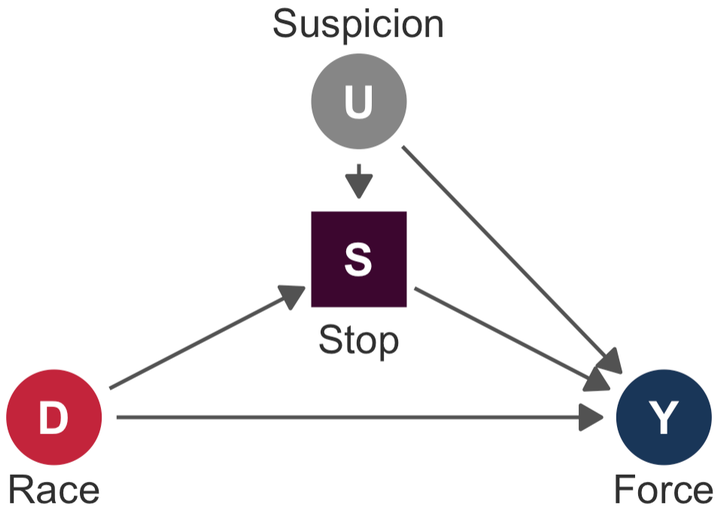
- **Race** (D) may affect whether someone is **stopped** (S), and whether **force** is used (Y)
- **Suspicion or circumstances** (U), unobserved, affect both stopping and force



Knox, Lowe & Mummolo (2020), *APSR* 114(3)

Draw the data-generating process as a DAG

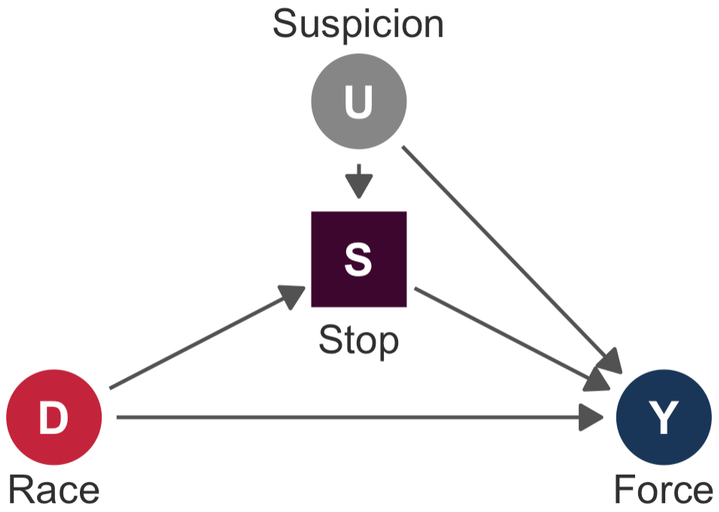
- **Race** (D) may affect whether someone is **stopped** (S), and whether **force** is used (Y)
- **Suspicion or circumstances** (U), unobserved, affect both stopping and force
- Records contain **only** $S = 1$: every analysis of records **conditions on** S



Knox, Lowe & Mummolo (2020), *APSR* 114(3)

Draw the data-generating process as a DAG

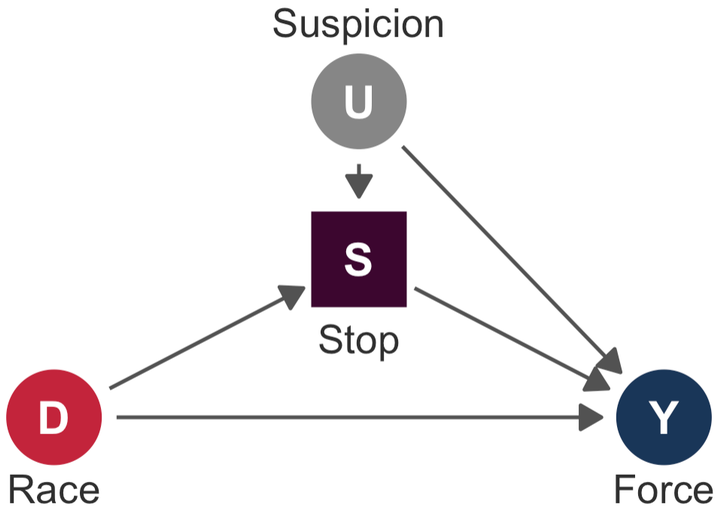
- **Race** (D) may affect whether someone is **stopped** (S), and whether **force** is used (Y)
- **Suspicion or circumstances** (U), unobserved, affect both stopping and force
- Records contain **only** $S = 1$: every analysis of records **conditions on S**
- S is a collider on $D \rightarrow S \leftarrow U \rightarrow Y$: conditioning on it opens a path that has nothing to do with how force is used



Knox, Lowe & Mummolo (2020), *APSR* 114(3)

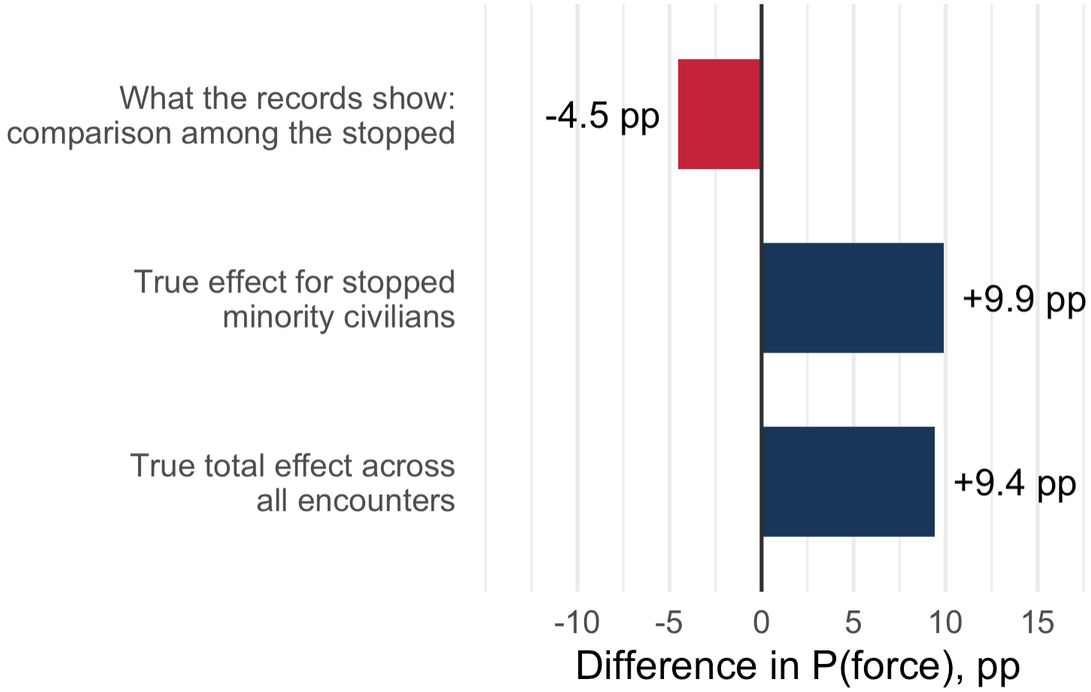
Draw the data-generating process as a DAG

- **Race** (D) may affect whether someone is **stopped** (S), and whether **force** is used (Y)
- **Suspicion or circumstances** (U), unobserved, affect both stopping and force
- Records contain **only** $S = 1$: every analysis of records **conditions on S**
- S is a collider on $D \rightarrow S \leftarrow U \rightarrow Y$: conditioning on it opens a path that has nothing to do with how force is used
- And the stop is also a **mediator** ($D \rightarrow S \rightarrow Y$): even the question "what is the effect?" needs care



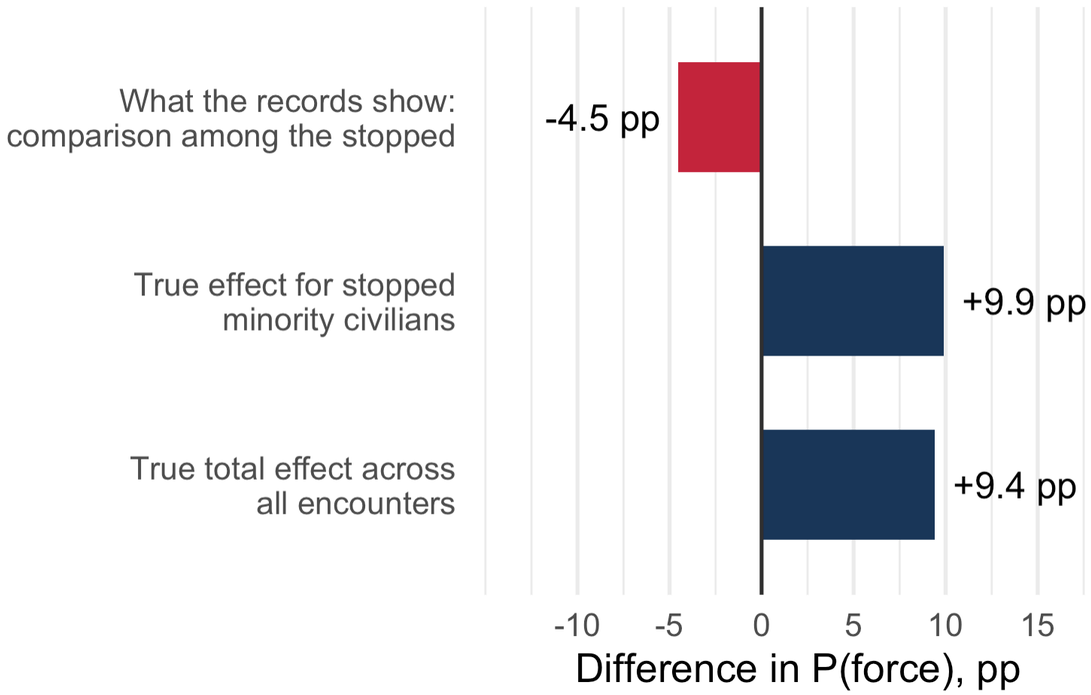
Knox, Lowe & Mummolo (2020), *APSR* 114(3)

How selection can hide discrimination



How selection can hide discrimination

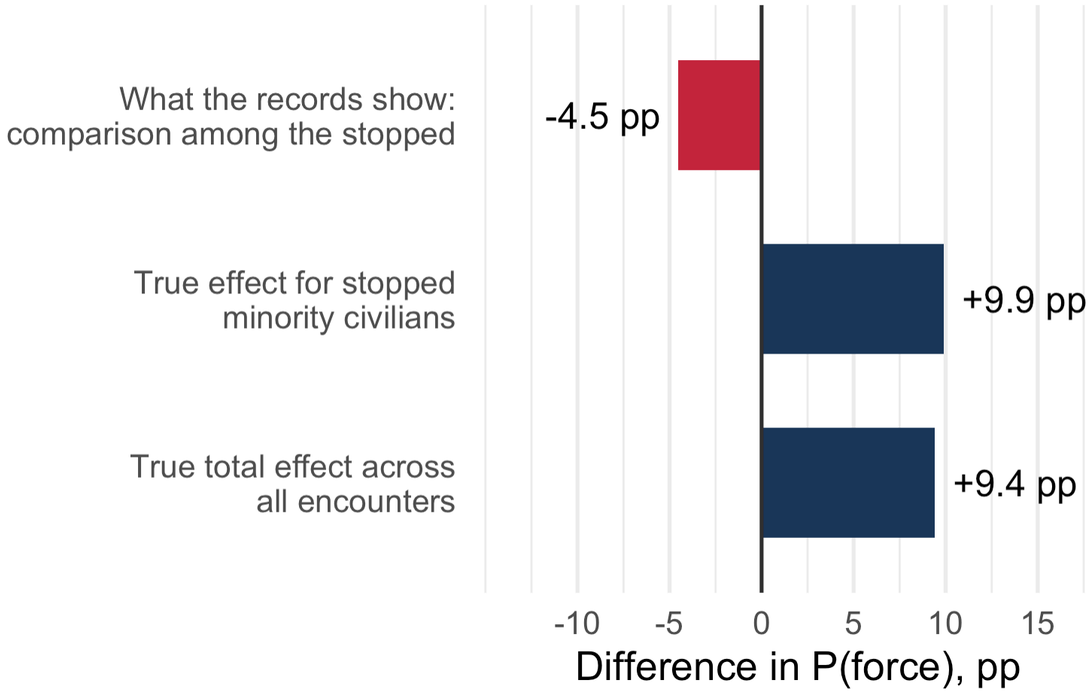
I simulated a city where officers stop minority civilians at **lower** levels of suspicion (i.e., lower trheshold for a stop), and once stopped, are more likely to use force on a minority civilian than on a white civilian with the same suspicion.



How selection can hide discrimination

> I simulated a city where officers stop minority civilians at **lower** levels of suspicion (i.e., lower trheshold for a stop), and once stopped, are more likely to use force on a minority civilian than on a white civilian with the same suspicion.

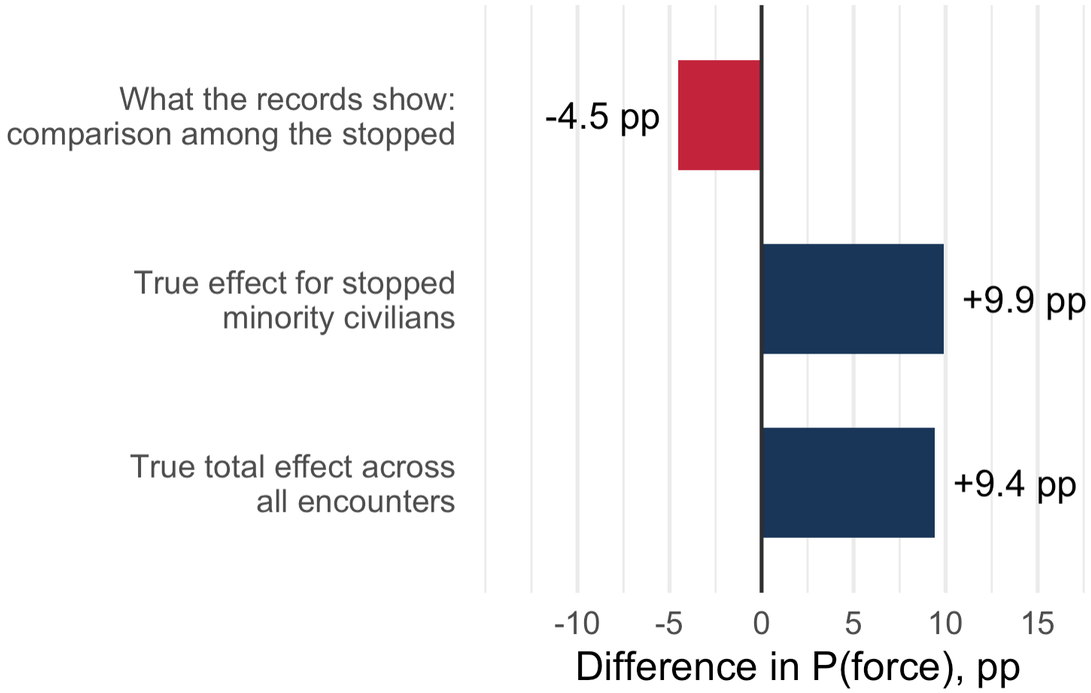
- In the records, minority civilians face force **4.5 pp less** often



How selection can hide discrimination

> I simulated a city where officers stop minority civilians at **lower** levels of suspicion (i.e., lower trheshold for a stop), and once stopped, are more likely to use force on a minority civilian than on a white civilian with the same suspicion.

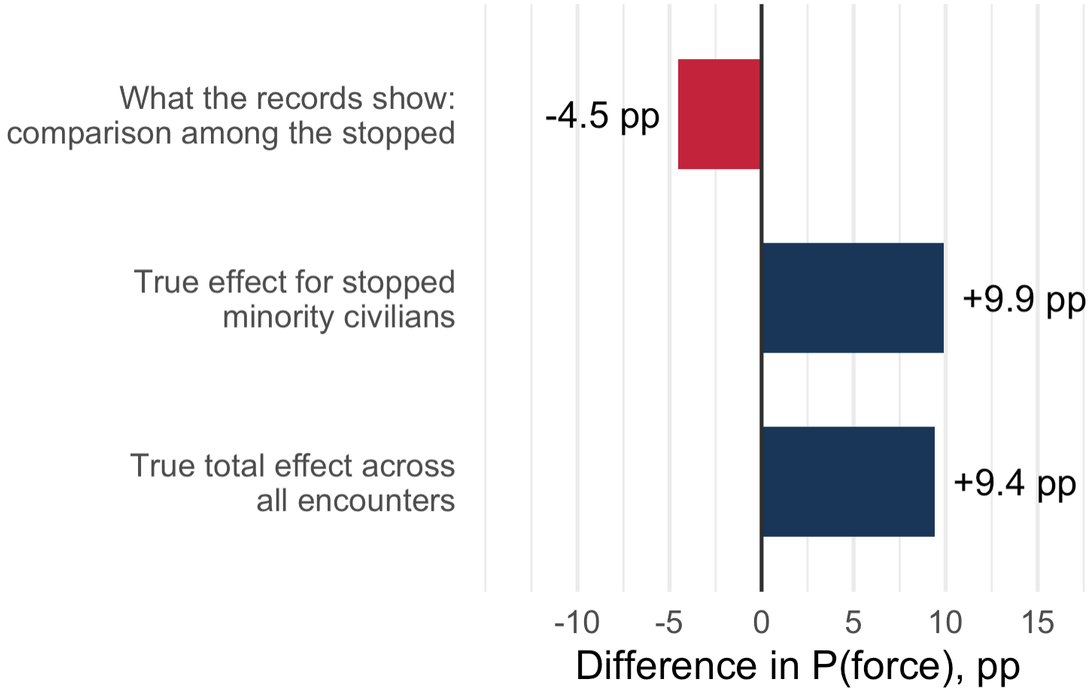
- In the records, minority civilians face force **4.5 pp less** often
- Yet the true effect is **+9.9 pp** for the same people



How selection can hide discrimination

> I simulated a city where officers stop minority civilians at **lower** levels of suspicion (i.e., lower trheshold for a stop), and once stopped, are more likely to use force on a minority civilian than on a white civilian with the same suspicion.

- In the records, minority civilians face force **4.5 pp less** often
- Yet the true effect is **+9.9 pp** for the same people
- Stopped minority civilians were stopped on **weaker suspicion**, so they look "less dangerous" in the records: the comparison **masks the discrimination entirely**, and a "no difference" finding is no evidence of no bias



Which estimand? Three candidates

Which estimand? Three candidates

Total effect, all encounters

How much does a civilian's race change the chance that an encounter ends in force?

Needs data on encounters that were never recorded

Which estimand? Three candidates

Total effect, all encounters

How much does a civilian's race change the chance that an encounter ends in force?

Needs data on encounters that were never recorded

Difference among the stopped

Among people who were stopped, how does force differ by race?

What records estimate: a description of a selected group, not the effect of race

Which estimand? Three candidates

Total effect, all encounters

How much does a civilian's race change the chance that an encounter ends in force?

Needs data on encounters that were never recorded

Difference among the stopped

Among people who were stopped, how does force differ by race?

What records estimate: a description of a selected group, not the effect of race

Effect among those stopped regardless of race

For civilians police would stop whatever their race, does race change the use of force?

Not identified from records without strong assumptions

Which estimand? Three candidates

Total effect, all encounters

How much does a civilian's race change the chance that an encounter ends in force?

Needs data on encounters that were never recorded

Difference among the stopped

Among people who were stopped, how does force differ by race?

What records estimate: a description of a selected group, not the effect of race

Effect among those stopped regardless of race

For civilians police would stop whatever their race, does race change the use of force?

Not identified from records without strong assumptions

The data *can probably* answer the middle question; the public (*and policy*) debate ask the first or the third.

Name the estimand first.

What Knox, Lowe & Mummolo do

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

What Knox, Lowe & Mummolo do

- Formalize the problem with potential outcomes and mediation: the stop is **both a mediator and a selection rule**

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

What Knox, Lowe & Mummolo do

- Formalize the problem with potential outcomes and mediation: the stop is **both a mediator and a selection rule**
- Derive the exact bias of the standard approach, and show that in most cases **no set of controls can remove it**

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

What Knox, Lowe & Mummolo do

- Formalize the problem with potential outcomes and mediation: the stop is **both a mediator and a selection rule**
- Derive the exact bias of the standard approach, and show that in most cases **no set of controls can remove it**
- Offer a bias correction and **nonparametric sharp bounds** for race effects, under explicit assumptions

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

What Knox, Lowe & Mummolo do

- Formalize the problem with potential outcomes and mediation: the stop is **both a mediator and a selection rule**
- Derive the exact bias of the standard approach, and show that in most cases **no set of controls can remove it**
- Offer a bias correction and **nonparametric sharp bounds** for race effects, under explicit assumptions
- Re-analyzing Fryer's (2019) NYPD data: traditional estimates can **severely understate** racially biased policing, or **mask it entirely**

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

What Knox, Lowe & Mummolo do

- Formalize the problem with potential outcomes and mediation: the stop is **both a mediator and a selection rule**
- Derive the exact bias of the standard approach, and show that in most cases **no set of controls can remove it**
- Offer a bias correction and **nonparametric sharp bounds** for race effects, under explicit assumptions
- Re-analyzing Fryer's (2019) NYPD data: traditional estimates can **severely understate** racially biased policing, or **mask it entirely**
- Propose a better design: record **encounters**, not only stops

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

What Knox, Lowe & Mummolo do

- Formalize the problem with potential outcomes and mediation: the stop is **both a mediator and a selection rule**
- Derive the exact bias of the standard approach, and show that in most cases **no set of controls can remove it**
- Offer a bias correction and **nonparametric sharp bounds** for race effects, under explicit assumptions
- Re-analyzing Fryer's (2019) NYPD data: traditional estimates can **severely understate** racially biased policing, or **mask it entirely**
- Propose a better design: record **encounters**, not only stops

> Bronner (2020) gives the plain-language version: statistics on police interactions describe only the small, non-random subset of people police chose to engage, so the relationships in those data can shift. A collider, in disguise.

Knox, Lowe & Mummolo (2020), *APSR* 114(3): 619–637 · Bronner (2020), FiveThirtyEight

Thinking with DAGs about how data are generated

> Before analyzing any dataset, draw how a row comes to exist:

Thinking with DAGs about how data are generated

> Before analyzing any dataset, draw how a row comes to exist:

- **Who could have entered the data?** The population before any selection: all encounters, all applicants, all crashes

Thinking with DAGs about how data are generated

> Before analyzing any dataset, draw how a row comes to exist:

- **Who could have entered the data?** The population before any selection: all encounters, all applicants, all crashes
- **Who decided whether they entered?** Officers, judges, editors, reviewers, survey non-response, platforms

Thinking with DAGs about how data are generated

> Before analyzing any dataset, draw how a row comes to exist:

- **Who could have entered the data?** The population before any selection: all encounters, all applicants, all crashes
- **Who decided whether they entered?** Officers, judges, editors, reviewers, survey non-response, platforms
- **Could that decision depend on your treatment or your outcome?** If on both, you are conditioning on a collider

Thinking with DAGs about how data are generated

> Before analyzing any dataset, draw how a row comes to exist:

- **Who could have entered the data?** The population before any selection: all encounters, all applicants, all crashes
- **Who decided whether they entered?** Officers, judges, editors, reviewers, survey non-response, platforms
- **Could that decision depend on your treatment or your outcome?** If on both, you are conditioning on a collider
- **What estimand do the data you have actually support?** Often a narrower one than your question

Thinking with DAGs about how data are generated

Before analyzing any dataset, draw how a row comes to exist:

- **Who could have entered the data?** The population before any selection: all encounters, all applicants, all crashes
- **Who decided whether they entered?** Officers, judges, editors, reviewers, survey non-response, platforms
- **Could that decision depend on your treatment or your outcome?** If on both, you are conditioning on a collider
- **What estimand do the data you have actually support?** Often a narrower one than your question

Previous research is part of the data-generating process too. This is a bit of a pet peeve for Kevin and me: published findings are selected on statistical significance (Think of Week 1). The literature you read is a selected sample…

IF TIME ALLOWS

Your turn 💬 (~5 min, groups of 3, then share)

Pick **one** ‘dataset’ (your own, or one below) and draw how a row comes to exist.

1. Asylum decisions in court records
2. Police-reported bicycle crashes (do protected lanes make cycling safer?)
3. Hospital admissions during a heatwave
4. Published studies on intergroup contact

IF TIME ALLOWS

Your turn 💬 (~5 min, groups of 3, then share)

Pick **one** ‘dataset’ (your own, or one below) and draw how a row comes to exist.

1. Asylum decisions in court records
2. Police-reported bicycle crashes (do protected lanes make cycling safer?)
3. Hospital admissions during a heatwave
4. Published studies on intergroup contact

Where is the selection node? Is it a collider between your treatment and outcome? What estimand do your data actually support?

When no adjustment set works, change the design

When no adjustment set works, change the design

- Knox et al.'s conclusion generalizes: when bias comes from **how the data were generated**, adding controls cannot fix it

When no adjustment set works, change the design

- Knox et al.'s conclusion generalizes: when bias comes from **how the data were generated**, adding controls cannot fix it
- Design-based strategies build comparisons that do not rely on measuring every confounder: **experiments**, **instrumental variables**, **regression discontinuity**, **difference-in-differences**

When no adjustment set works, change the design

- Knox et al.'s conclusion generalizes: when bias comes from **how the data were generated**, adding controls cannot fix it
- Design-based strategies build comparisons that do not rely on measuring every confounder: **experiments**, **instrumental variables**, **regression discontinuity**, **difference-in-differences**

COURSE

Causal Inference

The full toolkit of design-based strategies, offered as its own course

When no adjustment set works, change the design

- Knox et al.'s conclusion generalizes: when bias comes from **how the data were generated**, adding controls cannot fix it
- Design-based strategies build comparisons that do not rely on measuring every confounder: **experiments**, **instrumental variables**, **regression discontinuity**, **difference-in-differences**

COURSE

Causal Inference

The full toolkit of design-based strategies, offered as its own course

NEXT SPRING

Advanced Causal Topics

With me, spring 2027

When no adjustment set works, change the design

- Knox et al.'s conclusion generalizes: when bias comes from **how the data were generated**, adding controls cannot fix it
- Design-based strategies build comparisons that do not rely on measuring every confounder: **experiments**, **instrumental variables**, **regression discontinuity**, **difference-in-differences**

COURSE

Causal Inference

The full toolkit of design-based strategies, offered as its own course

NEXT SPRING

Advanced Causal Topics

With me, spring 2027

LAST SPRING

Advances in Diff-in-Diff

Slides and labs from the spring 2026 workshop: ask me for the materials

Wrapping up

Three questions before you run your regression

> A pre-going crazy with R (or whatever statistical software you use) checklist for any analysis that claims an effect (based on today's content):

Three questions before you run your regression

> A pre-going crazy with `R` (or whatever statistical software you use) checklist for any analysis that claims an effect (based on today's content):

- **What is the estimand?** Name the unit-specific quantity and the target population, and say whether it is descriptive or causal, before writing `lm()`

Three questions before you run your regression

> A pre-going crazy with R (or whatever statistical software you use) checklist for any analysis that claims an effect (based on today's content):

- **What is the estimand?** Name the unit-specific quantity and the target population, and say whether it is descriptive or causal, before writing `lm()`
- **Which DAG justifies the controls?** Every open backdoor path closed, and no new path opened by conditioning on a collider

Three questions before you run your regression

> A pre-going crazy with R (or whatever statistical software you use) checklist for any analysis that claims an effect (based on today's content):

- **What is the estimand?** Name the unit-specific quantity and the target population, and say whether it is descriptive or causal, before writing `lm()`
- **Which DAG justifies the controls?** Every open backdoor path closed, and no new path opened by conditioning on a collider
- **Is any control a mediator, a collider, or a selection rule?** If so, it does not belong in a model for the total effect, even if it "improves fit"

Three questions before you run your regression

A pre-going crazy with R (or whatever statistical software you use) checklist for any analysis that claims an effect (based on today's content):

- **What is the estimand?** Name the unit-specific quantity and the target population, and say whether it is descriptive or causal, before writing `lm()`
- **Which DAG justifies the controls?** Every open backdoor path closed, and no new path opened by conditioning on a collider
- **Is any control a mediator, a collider, or a selection rule?** If so, it does not belong in a model for the total effect, even if it "improves fit"

If you cannot answer the first question, the coefficient has no interpretation yet. Choosing controls is a **theoretical** decision, not a statistical one.

What we covered today

FROM QUESTION TO DATA

Causation and counterfactuals

Potential outcomes; ATE, ATT and ATC; the naive comparison = ATE + baseline bias + differential treatment effect bias

Estimands

Unit-specific quantity + target population, stated outside any model; descriptive targets are legitimate too

DAGs and controls

Conditioning on a confounder or mediator closes a path; conditioning on a collider opens one; choose controls from the graph, not from the data

Where data come from

Selection into the data is conditioning; administrative records can mask the effects we care about (Knox, Lowe & Mummolo)

Where this goes next

Where this goes next

WEEK 4 · MEASUREMENT I

We never see Y itself

Today we treated 'outcomes' as numbers in a table. In practice we observe a measure of it, Y*, not Y. What does measurement error do to an estimate, and does a finding travel to other people, places, and times?

Where this goes next

WEEK 4 · MEASUREMENT I

We never see Y itself

Today we treated 'outcomes' as numbers in a table. In practice we observe a measure of it, Y*, not Y. What does measurement error do to an estimate, and does a finding travel to other people, places, and times?

WEEK 5 · MEASUREMENT II

Constructs behind many answers

Sometimes 'outcomes' are not one survey answer but a latent construct behind many. How to measure it with a model, carry its uncertainty forward, and check that it means the same thing in every group we compare.

Readings

Compulsory

- Lundberg, Johnson & Stewart (2021). What is your estimand? *American Sociological Review* 86(3): 532–565.
- Cinelli, Forney & Pearl (2024). A crash course in good and bad controls. *Sociological Methods & Research* 53(3): 1071–1104.
- Knox, Lowe & Mummolo (2020). Administrative records mask racially biased policing. *APSR* 114(3): 619–637.
- Bronner (2020). Why statistics don't capture the full extent of the systemic bias in policing. *FiveThirtyEight*.

Optional

- Cunningham (2021). *Causal Inference: The Mixtape*, Ch. 3–4.
- Elwert & Winship (2014). Endogenous selection bias. *Annual Review of Sociology* 40: 31–53.
- Imbens (2020). Potential outcome and directed acyclic graph approaches to causality. *JEL* 58(4): 1129–1179.

Also cited today

- Behaghel, Crépon & Le Barbanchon (2015). *AEJ: Applied* 7(3): 1–27.
- Boisjoly, Duncan, Kremer, Levy & Eccles (2006). *AER* 96(5): 1890–1905.
- Fancourt & Steptoe (2019). *BMJ* 367: l6377.
- Fryer (2019). *Journal of Political Economy* 127(3): 1210–1261.
- Holland (1986). *JASA* 81(396): 945–960.
- Montgomery, Nyhan & Torres (2018). *AJPS* 62(3): 760–775.

Your regression returns a number.

Your estimand and your DAG decide

what that number is allowed to mean.

Next week: Measurement I · from concepts to numbers

In the lab on Monday

with Martín

In Lab 3 you'll draw today's DAGs in R, recompute the eight-student table, and simulate what confounders and colliders do to a regression.

`ggplot()` `dagify()` `paths()` `adjustmentSets()` `lm()`

LAB 3

`03-dags-lab.qmd`